Which Leaderboard Orderings Survive a Human Reader?

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Abstract

A leaderboard score for a free-text benchmark depends on two things: the model, and the *verifier*, the program that finds the answer in each response and checks it against the reference answer. Leaderboards report the first and not the second. On the Open LLM Leaderboard's own records, changing the verifier while keeping every response fixed reversed 15.9% of the pairwise orderings among 3,366 models (ties excluded). We ask which orderings a human reader would actually support.

On the archived MATH-Hard board, three verifiers read all 35,748 responses of 27 models, among them the leaderboard's 2024 verifier, whose published verdicts we reproduce on all 35,748. They disagree on 1,159 responses, and on 891 of these a verifier that gave no credit had found no answer it could read. Combining the verdicts with a human audit of 400 responses, we bound every accuracy gap and call an ordering *certified* when its bound excludes zero, with a finite-sample guarantee that covers all 351 orderings at once.

The audit alone certifies 0 of the 351 orderings: spread over 27 models, 400 labels cannot settle any ordering without an assumption that links models. If the rate at which a reader overrules three agreeing verifiers differs between models by at most ε, 174 orderings are certified at $\varepsilon = 0.01$ and 208 at $\varepsilon = 0$, and each ordering carries a breakdown value: the variation between models that would overturn it. A published ranking should name its verifier and report how many responses that verifier could not read.

Keywords: benchmark evaluation, answer verification, human audit, partial identification, leaderboards

1 Introduction

Public leaderboards shape which language models people download, fine-tune and deploy. For benchmarks whose answers are free text, such as competition mathematics, a score comes from two steps. The model writes a response, and then a program called a *verifier* reads it, finds the answer it gives, and checks that answer against the benchmark's reference, or *gold*, answer. The leaderboard reports the model's name and its score. It does not report which verifier produced the score, even though verifiers differ between evaluation tools, between tasks inside one tool, and over time for a single task.

The choice is not a detail. In February 2025 the Open LLM Leaderboard re-scored its MATH results with a different verifier (Kydlíček et al., 2025). We traced the change file by file through the leaderboard's own records: it rewrote the scores of runs it had already stored rather than generating new responses (Appendix G). Among the 3,366 models whose scores before and after come from the same run, 15.9% of the pairs swapped order (ties

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excluded). The responses were the same, so a ranking that moves this much is partly a ranking of verifiers.

A single response shows how this happens. Asked how many more euros than pounds a person called Kathy has, one model converted both amounts, compared them, and ended with: "Kathy has $\boxed{22}$ more euros than pounds. Final Answer: The final answer is 22. Great job!" The gold answer is 22, and a person reading the response would mark it correct. The leaderboard's verifier, however, looks for the sentence "Final Answer: The final answer is X. I hope it is correct." This response ended with "Great job!" where the verifier expected "I hope it is correct.", so the verifier found no answer and scored the response as wrong. A response in which no answer is found is recorded exactly like a response with a wrong answer, and the published score does not say which of the two happened. Whether a model follows the demonstrated format is a legitimate thing to measure; our question is narrower: what does a ranking say about the answers a reader would find?

Neither verifier is the truth, so a change of verifier shows that a ranking can move without saying which orderings to trust. A human reader is the reference, but reading every response of every model is rarely practical, and we are not aware of a measurement of how much of a published ranking a reader would support. We make that measurement on the archived MATH-Hard board of the Open LLM Leaderboard (Beeching et al., 2023; Fourrier et al., 2024), which published every model's response to each of its 1,324 competition problems. Three verifiers with different extraction rules read all 35,748 responses of 27 models. Where all three agree, a human reader almost always agrees with them; where they disagree, at least one of them is wrong. Most disagreements are not judgement calls: on 891 of the 1,159 disputed responses, a verifier that gave no credit had found no answer at all.

We then ask what a human audit of 400 responses can establish about the ranking. For each pair of models we find every accuracy gap consistent with the verdicts and the labels; if all of them favour the same model, the ordering is *certified*. The answer, with a guarantee that holds for all 351 orderings at once and needs no large-sample approximation, comes in two parts. On its own, the audit certifies 0 orderings: spread over 27 models, 400 labels leave every model's accuracy too loosely pinned down. Letting models share labels where all three verifiers agree changes this, at the price of an assumption that we state with a single number: the rate at which a reader overrules three agreeing verifiers may differ between models by at most a tolerance ε. The audit then certifies 174 orderings at a reference tolerance,[1] $\varepsilon = 0.01$, and 208 at $\varepsilon = 0$. Because each count rests on the assumption, we report the whole curve over ε and, for every ordering, a breakdown value: the variation between models it would take to overturn it.

The paper makes four contributions.

1. *The verifier moves the whole board.* On unchanged responses, the leaderboard's own change of verifier reversed 15.9% of the pairwise orderings among 3,366 models, and the scoring design behind this is common in four evaluation frameworks (Section 6).

1. Our analysis plan is a dated, version-controlled document released with the code, not a registration with an external registry, and it was written after the 400-item audit's labels existed. The reference tolerance is a point it fixes on a curve, not a confirmatory threshold; Appendix Table 6 orders each choice in the plan against the labels.

2. *Which verifier scored the board is established, not guessed.* Re-running the scoring code recorded in its result files reproduces the published verdict on 35,748 of 35,748 responses (Section 2), and most disagreements between verifiers are failures to find an answer (Section 3).

3. *A small audit settles nothing on its own.* This audit certifies no ordering without an assumption that links models, and every model's rank interval spans all 27 places (Sections 4 and 5).

4. *Principal result.* Under one stated assumption, how much of a ranking survives a human reader can be bounded and reported as a curve, with a breakdown value for each ordering (Section 5).

Section 2 describes the data, the verifiers and the human audit; Section 3 shows what the verifiers disagree about; Section 4 develops the method; Section 5 gives the answer; Section 6 looks beyond the roster; Sections 8 and 9 cover related work and limitations. We make no claim about models outside the roster or about benchmarks other than MATH-Hard.

2 The board, the verifiers, and the human audit

Everything in this paper uses one fixed set of model responses that the leaderboard published. This section describes those responses, the three verifiers that read them, how we sort the responses by what the verifiers say, and the small human audit that tells us what a careful reader would say.

2.1 The board and the roster

The Open LLM Leaderboard (Beeching et al., 2023; Fourrier et al., 2024) tested submitted models on MATH-Hard, the 1,324 hardest (Level 5) problems of the MATH benchmark of competition mathematics (Hendrycks et al., 2021), spread over seven subjects, and it published every model's written response to every problem. The leaderboard has since been retired, which suits this study: its responses can no longer change, and its scoring is on record. We call its table of models the *board*.

Our *roster* is the 27 models we study, a small part of a board that held 3,751 models at the February 2025 re-scoring (Kydlíček et al., 2025). The roster was not chosen for this paper. An earlier study ranked all 3,996 of the leaderboard's per-model result repositories by download count and kept the top 60, at most 4 per organisation, under a rule written down before any re-scoring; the roster is the 27 of these that received at least one problem in the human audit described below (Appendix F). It leans towards popular models, is neither complete nor random, and sits low on the board: the median roster model is at the 18th percentile of displayed accuracy (Appendix G). Every roster model answered every problem, so we work with all 35,748 responses.

2.2 Three verifiers

A verifier gives each response a verdict of 1 (credit) or 0 (no credit) in two steps. *Extraction* looks for the answer in the text, and the *equivalence check* then decides whether that answer

matches the gold answer. Either step can fail, and a response whose answer cannot be found gets 0, exactly like a wrong answer. Our three verifiers differ mainly in how they extract.

The 2024 verifier is the leaderboard's original scorer, from the lm-evaluation-harness (Biderman et al., 2024). It reads an answer only from the sentence "Final Answer: The final answer is X. I hope it is correct.", which the prompt demonstrates and which we call the *template*, and it checks whether that answer is mathematically equal to the gold answer rather than identical as text. The leaderboard published this verifier's verdict on every response, together with the scoring code that ran. Re-running that code reproduces the published verdict on 35,748 of 35,748 responses, so we know exactly which verifier produced the original scores.

Math-Verify is the verifier the leaderboard switched to in February 2025 (Kydlíček et al., 2025). The switch changed the displayed scores but left the published per-response verdicts as they were, so from then on the board showed scores from one verifier and published verdicts from another. Running Math-Verify ourselves reproduces the re-scored totals in 168 of 175 model-subject combinations; each of the others is off by a single problem.

The boxed verifier reads the expression inside the last `\boxed{}` command, which is how the benchmark's own reference solutions mark their answers, and compares it in the same way. Appendix Table 4 gives each verifier's version and the check behind it.

The three are not independent. All rest on the same algebra library, and the 2024 and boxed verifiers also share the harness's normalisation of answers and the stored gold answer, which the harness takes from the same reference solution that Math-Verify reads (Appendix Table 5). Nothing below assumes independence; what matters is that the verifiers fail on different responses, and they do. While tracing the verdicts we also found that our own boxed verifier had been silently degraded on Windows, where an error handler turned every check of mathematical equality into a plain comparison of text. All results use the corrected version. Appendix F reports the defect, which is itself an instance of this paper's subject: a verifier that fails quietly still produces a plausible number.

2.3 Sorting responses by the verdicts

Writing the $K = 3$ verdicts in a fixed order (the 2024 verifier, the boxed verifier, Math-Verify) gives each response a three-digit string, which places it in one of 8 *cells*. Cell 111 holds the responses that all three verifiers credit and cell 000 those that none credits; in cell 101, for example, the 2024 verifier and Math-Verify credit the response and the boxed verifier does not. We call 111 and 000 the *concordant* cells, because there the verifiers agree, and the other six the *discordant* cells. Table 1 gives their sizes.

2.4 The human audit and the reader verdict

The quantity this paper is after is whether the answer a response gives is right, as a careful reader would judge it. A ranking can also legitimately reward following the requested format; we measure that separately in Section 3. We obtain the reader's judgement in two steps, and only the first is human. A labeller sees the problem and the response and nothing else: not the model, not any verifier's verdict, and not the gold answer. The labeller copies down, exactly as written, the final answer the response *commits to*, meaning the answer it presents as final after any corrections it makes along the way. If there is no such answer the

Table 1: The eight cells on MATH-Hard. A cell is the string of the three verifiers' verdicts in the order 2024 verifier, boxed verifier, Math-Verify; 1 is credit. The first two rows are the concordant cells, where all three verifiers agree, and only they enter the certification. The last column gives, for the 400-item audit, how many audited responses in the cell have reader verdict 1, out of how many were audited.

cell	verdicts	responses	reader verdict 1
000	none credits	32,685	0/223
111	all three credit	1,904	142/144
101	only 2024 and Math-Verify credit	448	–
011	only boxed and Math-Verify credit	349	16/33
001	only Math-Verify credits	273	–
100	only 2024 credits	40	–
110	only 2024 and boxed credit	34	–
010	only boxed credits	15	–

labeller writes NONE, and if the response gives two different final answers without choosing between them, CONTRADICTORY. The benchmark's own equivalence check then compares the copied answer with the gold answer. The result is the *reader verdict*, written Y^*: 1 if they match, and 0 otherwise, including for NONE and CONTRADICTORY (Appendix E drops those instead). Y^* is a measurement, not mathematical truth: it records which answer a reader finds and whether the benchmark accepts it.

All human labels come from one audit of 400 responses, designed for an earlier study in a dated document committed before labelling began. The labellers worked from a spreadsheet that showed only the problem and the response. Of the 400 labels, 144 fall in cell 111 and 223 in cell 000. Because the audit was drawn by the boxed verifier's verdict and by subject, not by model and cell, it has too few labels to say anything about one model in one discordant cell. We therefore use only its labels in the two concordant cells and treat every discordant cell as unknown, which can only make our conclusions more cautious. The author labelled all 400 responses; two other labellers labelled overlapping sets of 151 and 75, and using their labels instead moves the certified count by at most 2 (Appendix F). Our analysis plan also describes a larger audit drawn by model and cell and a complete labelling of the discordant responses; neither has been carried out, and no result here depends on them.

3 What the verifiers disagree about

Before any assumption or statistics, the verdicts themselves show where the verifiers part ways and why. Apart from one comparison with the human audit, nothing in this section uses a human label.

The verifiers agree on almost every response. Of the 35,748 responses, 1,904 are credited by all three verifiers and 32,685 by none (Table 1). Only the remaining 1,159 get a different verdict depending on the verifier. Where the three agree, the reader nearly always agrees with them: the reader verdict is 1 for 142 of the 144 audited responses in cell 111

Table 2: Why the verifiers disagree: every discordant response, classified from what each verifier's extraction step recorded, with no human labels. *Extraction*: a verifier that gave no credit found no answer. *Judgement*: every verifier that gave no credit extracted an answer. *Runtime*: a verifier timed out. The classes are exclusive.

class	responses
extraction	891
no boxed answer (boxed verifier finds nothing)	533
template sentence missing (2024 verifier finds nothing)	205
template+box	153
judgement	268
runtime	0
all discordant	1,159

and for 0 of the 223 in cell 000 (Appendix Figure 4). This is what will later let us share labels between models in those two cells.

Most disagreements are failures to find an answer. Table 2 sorts every discordant response by what each verifier's extraction step recorded, using fixed rules and no human labels (Appendix I). In 891 of the 1,159, a verifier that gave no credit had found no answer to judge: 533 responses contain no boxed answer, 205 do not use the template, and 153 have neither. In the other 268, every verifier that gave no credit did find an answer, and they disagree about which answer the response gives or whether it matches the gold answer. No disagreement comes from a verifier timing out. Appendix Table 8 shows one randomly drawn example of each kind.

Following the format and being right are different things. 23,400 of the 35,748 responses (65.5%) use the template, in the sense that the 2024 verifier's pattern finds an answer in them, and across models the share runs from 0.1% to 95.6% (Appendix Table 9). A published score mixes the two properties. Of the 637 responses that the boxed verifier or Math-Verify credits and the 2024 verifier does not, 358 skip the template; for those, the 2024 verifier's zero means it could not read the answer.

Neither strict nor lenient extraction is safe. The errors run both ways, which is why we do not simply pick a verifier. A strict rule like the 2024 verifier's misses correct answers that skip the template: among the 158 audited responses with reader verdict 1, 3 skip it, and the 2024 verifier credits 0 of them. A lenient rule like the boxed verifier's or Math-Verify's also finds answers a reader would not accept: among the 71 audited responses that skip the template and have reader verdict 0, the boxed verifier credits 17 and Math-Verify 17. The audit is not a random sample of the board, so these are counts, not rates.

A change of wording can flip a verdict. A check that needs no reader points the same way. We took 300 responses that all three verifiers credit and rewrote each in ways that leave the stated answer untouched and that occur in real responses. Replacing the template's closing words with "Great job!", as in the introduction's example, leaves the other two verifiers crediting every response and the 2024 verifier crediting 0. Removing the

template but keeping the box leaves the 2024 verifier crediting 0. Removing the box but keeping the template leaves the 2024 verifier crediting every response, the boxed verifier 0, and Math-Verify 299. Each verifier can be moved by a change of form that leaves the answer as it was.

So the verifiers disagree on a small share of responses, mostly because one of them could not read an answer, and how often that happens depends strongly on the model. Whether it is enough to change the ranking is the question the next two sections answer.

4 From a small audit to a ranking

This section turns the question "which orderings survive a human reader?" into something we can compute. We define a model's accuracy under the reader verdict, show why 400 labels cannot settle a ranking by themselves, state the one assumption that lets models share labels, and explain when an ordering counts as settled and with what confidence. Proofs are in Appendix A.

4.1 What an ordering claims

Each model's responses are spread over the eight cells. Let s_{mc} be the *share* of model m's responses that fall in cell c; shares are known exactly, because every verifier reads every response. Let p_{mc} be the fraction of those responses with reader verdict $Y^* = 1$. These fractions are what we do not know. Model m's accuracy under the reader verdict, and the gap between models j and k, are

$$A_m \;=\; \sum_c s_{mc}\, p_{mc}, \qquad \Delta_{jk} = A_j - A_k. \tag{1}$$

An *ordering* $j \succ k$ is the claim that $\Delta_{jk} > 0$: model j would score higher than model k if a reader had marked every response. The 27 models form 351 pairs, and for each pair we ask whether either ordering can be established.

The difficulty is coverage. The 400 labels leave most combinations of model and cell with no label at all, and for those the only safe statement is $p_{mc} \in [0, 1]$. Each such cell widens the range of possible gaps by its share, and when the widening exceeds the gap itself the ordering stays open. The discordant cells are small, so leaving them open costs little. The concordant cells hold almost every response, so they must be pinned down tightly, and that requires letting models share labels.

4.2 Letting models share labels

Sharing labels between models, which statisticians call *pooling*, is an assumption, not a measurement, and we state it with a single number.

Assumption 1 (Bounded heterogeneity, $H(\varepsilon)$)**.** For each concordant cell c, the values of p_{gc} across the audit sub-populations g that make up that cell lie in an interval of width at most ε.

In words: where all three verifiers agree, the rate at which a careful reader would overrule them may differ from model to model, but by no more than ε. A sub-population is usually

a model; where the audit split a model's responses into groups (by subject and by the boxed verifier's verdict), each group counts separately, which makes the assumption slightly stronger. The assumption says nothing about the overrule rate itself, which the audit estimates, only about how much it varies.

Setting $\varepsilon = 0$ is stronger than it looks. Each p_{gc} is a fraction of a finite set of responses, so two models with the same underlying chance of being overruled will still, by luck, end up with slightly different fractions. $H(0)$ is therefore a condition on the actual responses, not a consequence of equal rates. We report every result as a function of ε and mark $\varepsilon = 0.01$, fixed in our analysis plan as a reference. That value is a judgement, not a measurement, and it was set after the audit's labels existed (Appendix Table 6), which is why no count in this paper appears without the curve around it.

4.3 Certified orderings and breakdown values

The idea is simple. In each concordant cell the audit gives a confidence interval for the shared rate, and under the assumption every model's own rate lies within ε of it. In each discordant cell a model's rate may be anything. Taking the least and the most favourable values in every cell gives the smallest and the largest gap two models could have. Formally, let $\tilde{p}_c$ be the pooled fraction of responses with $Y^* = 1$ in concordant cell c, weighted as the audit sampled the sub-populations, and let $[L_c, U_c]$ be a confidence interval for it, built in Section 4.5 so that it does not assume the models are alike. For a discordant cell, let $[l_{mc}, u_{mc}]$ be an interval for model m alone, which is $[0, 1]$ when that model has no labels in the cell.

Proposition 1 (Identified range)**.** *Fix a pair (j, k) and a concordant cell c. Under Assumption 1 and on the event $\tilde{p}_c \in [L_c, U_c]$, the pair (p_{jc}, p_{kc}) lies in*

$$R_c(\varepsilon) \;=\; \big\{(x, y) \in [\ell_c, h_c]^2 \;:\; |x - y| \le \varepsilon\big\}, \qquad \ell_c = \max(0, L_c - \varepsilon), \quad h_c = \min(1, U_c + \varepsilon).$$

Consequently Δ_{jk} lies in the interval obtained by minimising and maximising Equation (1) separately in each cell, over $R_c(\varepsilon)$ for concordant cells and over the per-model intervals for discordant ones.

We call this interval the *identified range* of Δ_{jk}: every gap between the two models that is consistent with the verdicts, the labels and the assumption. Appendix A proves the proposition and solves the optimisation in closed form, and Appendix B checks both against brute force. At $\varepsilon = 0$ the range is exact; for $\varepsilon > 0$ it is slightly wider than necessary, which can only make conclusions more cautious.

An ordering $j \succ k$ is *certified* at ε when the whole identified range of Δ_{jk}, widened by any sampling term from Section 4.5, lies above zero. A larger ε gives a wider range, so an ordering certified at one tolerance is certified at every smaller one, which makes the following well defined.

Definition 2 (Breakdown value)**.** $\varepsilon^*_{jk} = \sup\{\varepsilon \in [0, 1] : j \succ k \text{ is certified at } \varepsilon\}$*, with* $\varepsilon^*_{jk} = 0$ *when the ordering is not certified at* $\varepsilon = 0$*.*

The breakdown value is how much model-to-model variation an ordering can absorb before it stops being certified. Reporting it for every ordering turns the dependence on

Assumption 1 into a number that a reader can hold against their own view. Certification is a strict inequality, so the supremum need not be attained, and because the range is cautious for $\varepsilon > 0$, each reported value understates the true one.

4.4 Why three verifiers rather than one

Proposition 3 (A common error mechanism cannot reorder)**.** *Suppose one verifier is used, and suppose its predictive values are common to all models, so that* $\Pr(Y^* = 1 \mid v = 1) = \pi_1$ *and* $\Pr(Y^* = 1 \mid v = 0) = \pi_0$ *regardless of* m*. Then* $\Delta_{jk} = (\pi_1 - \pi_0)(a_j - a_k)$*, where* a_m *is the rate at which the verifier credits model* m*.*

Every gap is scaled by the same factor, so a verifier whose mistakes are alike for all models cannot swap one pair while leaving another in place. The swaps that Section 6 reports therefore need mistakes that differ between models, and correcting a single verifier with error rates shared by all models would assume away exactly what we study. Sharing labels only where verifiers with different extraction rules agree leaves the disagreement cells free to behave differently for each model.

4.5 Confidence

An interval that does not assume the models are alike. The pooled count in a concordant cell adds up labels from models whose true rates may differ, and the labels were drawn without replacement within each stratum of the audit. A textbook binomial interval assumes one common rate and sampling with replacement; using it here would quietly build the sameness that Assumption 1 makes explicit into the confidence statement. We use a bound that holds without either condition.

Lemma 4 (Design-valid pooled bound)**.** *Let* x_c *be the total number of audited responses with* $Y^* = 1$ *in cell* c*, summed over sub-populations, with* n_c *labels in total and sampling weights* $w_{gc} = n_{gc}/n_c$*. Let* $\tilde{p}_c = \sum_g w_{gc}\, p_{gc}$*. Then, conditional on the realised number of labels in each sub-population, for* $0 < t \le 1 - \tilde{p}_c$ *and* $0 < t \le \tilde{p}_c$ *respectively (larger* t *make the events empty), with* KL *as defined in Appendix A,*

$$\Pr\left(\frac{x_c}{n_c} \ge \tilde{p}_c + t\right) \le e^{-n_c \mathrm{KL}(\tilde{p}_c + t \,\|\, \tilde{p}_c)}, \qquad \Pr\left(\frac{x_c}{n_c} \le \tilde{p}_c - t\right) \le e^{-n_c \mathrm{KL}(\tilde{p}_c - t \,\|\, \tilde{p}_c)}.$$

The proof uses two standard facts about moment generating functions: sampling without replacement is dominated by sampling with replacement, and a sum of independent Bernoulli variables with different means is dominated by a single Bernoulli at their weighted mean. Inverting the bound gives $[L_c, U_c]$ with at least nominal coverage in finite samples.

Two targets: these problems, and questions like them. The *primary target* is Δ_{jk} on the 1,324 problems the leaderboard actually used, because that is what a leaderboard reports: a score on a fixed set of questions. For this target the only uncertainty comes from the audit, the only assumption is Assumption 1, and the guarantee holds at the audit's actual size, with no large-sample approximation. Every headline number in the paper is of this kind. A reader may also want a claim about questions like these in general. MATH-Hard is a curated benchmark, not a random draw from a defined population, so such a claim has to

treat the problems as if they were one, which adds an uncertainty that no amount of labelling can remove. We bound it with a margin for item sampling (a paired half-width) built from quantities the identified range already supplies; it relies on a central limit approximation, and a distribution-free alternative is reported beside it (Appendix C). This *secondary target* is analysed in full, and its numbers are never mixed with the primary ones. The order of the two departs from our analysis plan, which first named the secondary target as its headline; we made the change after the audit's results for both targets were known and recorded it as an amendment to the plan (Appendix Table 6).

All orderings at once. Someone reading a leaderboard reads all 351 orderings together, so every headline count is simultaneous: the audit's error level is divided over every parameter on the roster (Appendix C), and for the secondary target the sampling term also receives a Bonferroni correction, which divides its error level among the orderings. Rank intervals follow directly from the certified set (Appendix A).

Proposition 5 (Coverage)**.** *Under Assumption 1, all claims about the primary target hold simultaneously with probability at least* $1 - \alpha_{\text{audit}}$ *at any sample size, with no large-sample approximation. Claims about the secondary target hold with probability at least* $1 - \alpha_{\text{audit}} - \alpha_{\text{sample}}$*, asymptotically in the number of items.*

The event behind the primary guarantee is that every concordant interval covers its target, and that event does not involve ε. On it, the orderings certified at every tolerance for which Assumption 1 holds are all correct, so the whole primary curve can be read at once, with no further correction for looking at many tolerances.

A simulation at the roster's scale, with 2,000 replicates per scenario under the real design (Appendix C), checks whether the guarantee is attained. Where Assumption 1 holds, no ordering was wrongly certified in any scenario, which bounds the family-wise error, the chance that even one certified ordering is wrong, below 0.0015 at 95% confidence; with heterogeneity that the analysis does not allow for, the error rises to 0.0675, above the nominal 0.025. When every model shares one underlying rate and the analysis uses $\varepsilon = 0$, the two-verifier analysis errs at rate 0.185 and the three-verifier analysis at 0.017, and a tolerance of 0.01 removes the error in both. This is why we do not lead with $\varepsilon = 0$.

5 How many orderings survive

We now apply the method to the roster. The main answer concerns the 1,324 problems the leaderboard actually used; we then ask what changes if the problems are treated as a sample of possible questions, and how the answer depends on the verifiers and on our analysis choices.

5.1 On these problems: the principal result

Every count in this subsection holds for all 351 orderings at once: the probability that all of them are right together is at least 0.975, with no large-sample approximation, but only under Assumption 1 at the stated ε. With no sharing of labels between models, which assumes nothing about how the models relate, no ordering is certified: 0 of 351. Allowing the overrule rate to differ between models by at most $\varepsilon = 0.01$ certifies 174 orderings, and

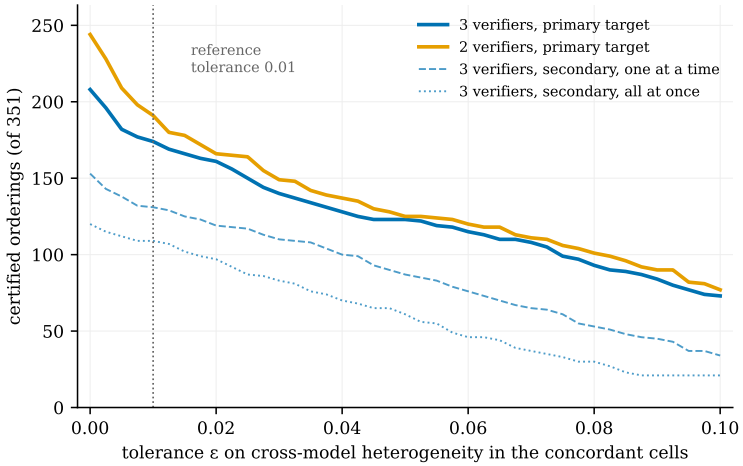


Figure 1: The paper's main result: certified orderings as a function of the tolerance ε, the most that the reader's overrule rate may differ between models where the verifiers agree. Solid lines are the primary target (these problems, all pairs at once) with three and with two verifiers. Thin blue lines are the secondary target with three verifiers, one ordering at a time (dashed) and all at once (dotted). The vertical line marks the reference tolerance.

requiring the models to be exactly alike, $\varepsilon = 0$, certifies 208. The answer to the paper's question is therefore a curve, not a single number.

Figure 1 draws that curve. The certified count falls smoothly as the tolerance grows. Among the orderings certified at $\varepsilon = 0$, the median breakdown value is $\varepsilon^* = 0.072$, several times the reference tolerance: half of these orderings would survive that much model-to-model variation in the concordant cells, and half would not.

Figure 2 shows what this means for someone reading the board. The printed ranking takes the 2024 verifier as right about every response and gives each model a single rank. Measured against a human reader, only 1 rank is pinned down at the reference tolerance: the top model keeps first place. The median model's rank is uncertain by 12 places at $\varepsilon = 0$ and by 17 at the reference tolerance, and without shared labels any model could hold any of the 27 places.

5.2 Beyond these problems

A claim about questions like these in general adds the item-sampling term of Section 4.5, and the counts fall. Even with the 2024 verifier taken as right about every response, item sampling alone allows only 280 orderings to be certified one at a time, which is the ceiling for this target. With the audit, 153 orderings are certified one at a time at $\varepsilon = 0$ and 131 at the reference tolerance, and 109 when all must hold at once; the distribution-free sampling term certifies 100 one at a time at $\varepsilon = 0$. These counts answer a different question at a different confidence level and are not compared with the primary ones (Appendix Figure 3).

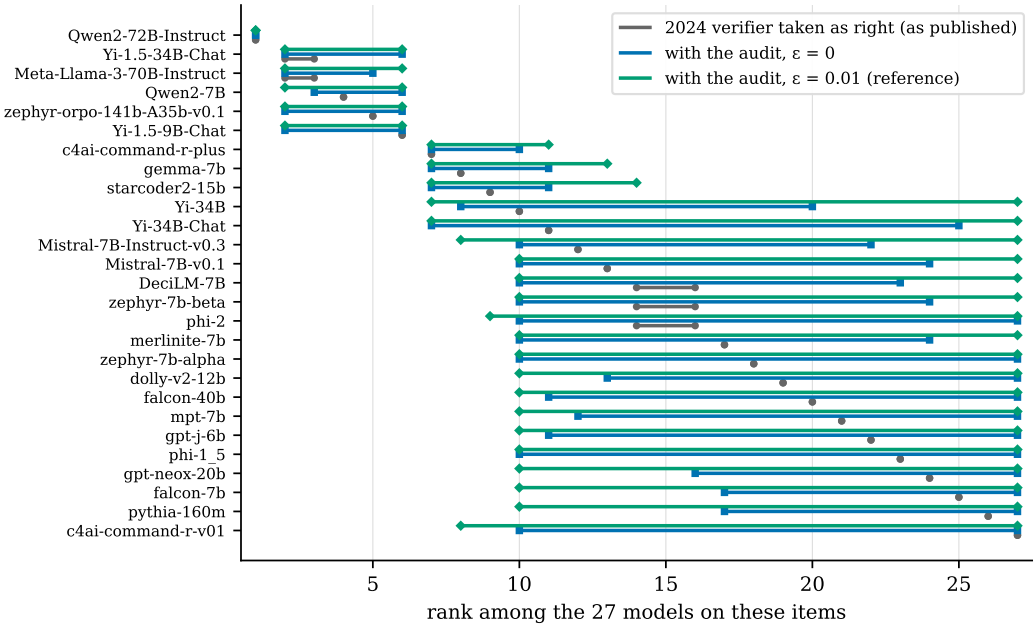


Figure 2: Each model's possible ranks on these problems, holding for all orderings at once with family-wise confidence 0.975. Grey is the ranking the leaderboard prints, which takes its 2024 verifier as right about every response and has no interval except where accuracies tie. Blue and green are our analysis at $\varepsilon = 0$ and at the reference tolerance. Without sharing labels between models (not drawn), every model's interval spans all 27 ranks.

5.3 How the answer depends on the verifiers and on our choices

Fewer verifiers certify more, and a fourth changes almost nothing. With only the leaderboard's two verifiers, 244 orderings are certified at $\varepsilon = 0$ and 191 at the reference tolerance, against 208 and 174 with three. Dropping the boxed verifier sends the responses on which only it disagrees back into the concordant cells, where they are pinned down tightly. The price is a stronger assumption: at $\varepsilon = 0$ the two-verifier analysis is the one whose error exceeded its nominal level in simulation. Because the three verifiers share machinery, we also added xFinder (Yu et al., 2025), a fine-tuned language model that returns the answer a response commits to or declines to give one. Its answers are judged by the same equivalence check as the reader verdict, so it differs from the other verifiers only in how it extracts. It credits 2,909 responses and declines on 9,092, which score 0, as an unfound answer does for the other verifiers (Appendix F). With four verifiers the discordant responses grow from 1,159 to 1,190, and the certified count at the reference tolerance moves from 174 to 173 (173 with xFinder's other system prompt). The verifiers do rank the models differently: across the four, 23 of the 27 models change rank, one by 17 places (Appendix Table 9).

Our own analysis choices move the count by a few orderings at most. At the reference tolerance, dropping NONE and CONTRADICTORY labels gives 174, Clopper–Pearson

intervals give 174, judging the copied answer with Math-Verify instead of the benchmark's check gives 172, and string comparison, which also changes the boxed verifier, gives 168 (Appendices D and E).

6 How common is this design choice?

The results so far concern one roster on one board. Two wider questions can be answered without new labels: whether the choice of verifier also moves the rest of the board, and whether the scoring design behind the problem is common elsewhere.

Across the whole board. The re-scoring, on 13 February 2025, rewrote the results file of a model's run with Math-Verify's scores and did not generate new responses. We traced every displayed score on the board back to its results file (Appendix G). For the 3,366 models whose displayed score comes from a file the re-scoring rewrote, and whose earlier version we could read, the file's score before and after the rewrite compares two verifiers on the same responses. Of the pairs that both scores order strictly, 15.9% reverse, that is, swap places (Appendix G defines the statistic and its treatment of ties), and Kendall's rank correlation between the two scores, which would be 1 if they ordered every pair alike, is $\tau_b = 0.67$. Among them, on the 25 roster models, 7.5% reverse ($\tau_b = 0.84$), so the roster understates how far the choice of verifier moves the board.

Across evaluation frameworks. The design choice at the root of the problem is to score an unreadable response as a wrong answer. We loaded the task configurations of four evaluation frameworks (Biderman et al., 2024; Cao et al., 2026; Habib et al., 2023; OpenAI, 2024) with each framework's own loader and classified the scoring path of each generative task. In `lm-evaluation-harness`, 2,936 of 4,524 generative tasks (64.9%) match an assumed output format and score a failure to match as incorrect. In OpenCompass the figure is 158 of 1,070 configurations, in `lighteval` 13 of 37 generative metrics, and in `simple-evals` 4 of 6 string-matching modules (Appendix Table 7). A behavioural check, run on a random sample of `lm-evaluation-harness` tasks only, confirmed every classification it could test (Appendix H).

These counts say how many tasks could be affected, not how many rankings change; that depends on how often real responses escape the extraction rule, which we have measured for one benchmark only.

7 Released data and code

Everything this paper reports can be regenerated from the release at `https://github.com/Rajveer-code/leaderboard-orderings`, and the release is also a resource in its own right: a fully adjudicated record of where three verifiers disagree on a public leaderboard, with human labels.

What is released. (i) The verdict of each verifier, including xFinder, on every one of the 35,748 responses of the 27 models, with the parse and time-out flags each verifier records; responses are identified by model and item identifier rather than copied. (ii) The human labels: the 400-item audit (the answer the labeller found and the resulting reader verdict) and the overlaps of 151 and 75 items labelled by two further labellers. (iii) The labelling

sheets, the offline labelling application and the rulebooks, including those of the larger audits that were designed but not labelled, which enter no result. (iv) The board records: for every model scored before the change of verifier, the results file behind its displayed score and that file's history (Appendix G), and the census of scoring rules in four evaluation frameworks. (v) The code that computes every number, table and figure from these files, with its tests, and the dated analysis plan.

Sources and licences. The problems and reference solutions are those of MATH-Hard as distributed for the leaderboard (`lighteval/MATH-Hard`, MIT licence). The model responses come from the leaderboard's public per-model details datasets on the Hugging Face Hub, whose dataset cards declare no licence; the release therefore copies the text only of the responses shown to labellers, for research use, and refers to every other response by model and item identifier, which match the public datasets. Our code, verdicts and labels are released under the Apache 2.0 licence.

Format, hosting and maintenance. Verdicts and labels are CSV and JSON files with one row per response. The release is hosted on GitHub and maintained by the author, who will answer issues there; every change is recorded in the repository's history, and the version that produced this paper is tagged `dmlr-submission`. A response's verdicts depend on each verifier's version, and the release pins the versions of the libraries it uses.

Responsible use. The labels describe the answers that model responses to competition mathematics problems give; they contain no personal data and no information about the labellers. They measure what a careful reader finds in a response, not whether a model followed the demonstrated output format, and they should not be used to rank models outside the roster without a new audit.

8 Related work

Appendix J adds two-phase sampling, verification bias, raters without a gold standard, uncertainty in rankings, design sensitivity and stratified allocation.

Whether a benchmark measures what it claims. BetterBench grades benchmarks against criteria for design, implementation and documentation (Reuel et al., 2024), a complementary framework asks which evaluative claims a piece of evidence supports (Salaudeen et al., 2025), and retrieval tools find the items in a benchmark that bear on a particular use (Diddee et al., 2026). That work sits upstream of ours: we take the benchmark and its gold answers as given and ask whether the published ranking is determined by the pipeline's own outputs, which a valid benchmark does not guarantee.

Configuration moves scores, and extraction is part of the configuration. Evaluation results move with prompt formatting (Sclar et al., 2023), with small changes that reorder leaderboards (Alzahrani et al., 2024), and with the many settings of a widely used evaluation harness, answer extraction among them (Biderman et al., 2024). The sentence that the 2024 verifier looks for descends from the convention introduced with Minerva (Lewkowycz et al., 2022), and learned extractors have followed (Yu et al., 2025). The closest study on the extraction side concerns multiple-choice evaluation, where extraction methods can disagree with human judgement and produce misleading model comparisons (Molfese

et al., 2025). We study free-form mathematical answers and ask which pairwise comparisons can be established when no extraction method is taken as correct. Benchmarks of verifier accuracy (Li et al., 2025) and verifiers trained to handle the formats that defeat regular expressions (Liu et al., 2025) go the other way: rank the verifiers and adopt the best. A better verifier shrinks the discordant cells but does not settle the responses on which verifiers still disagree.

Label-efficient evaluation with a proxy. Prediction-powered inference combines many machine-generated labels with a small human-labelled sample to produce valid confidence intervals (Angelopoulos et al., 2023), with variants for hybrid evaluation of language models (Fisch et al., 2024), active labelling (Zrnic and Candès, 2024) and synthetic labels (Boyeau et al., 2024). Two papers target rankings: Chatzi et al. (2024) return rank sets with a coverage guarantee from pairwise comparisons by a strong model calibrated against human comparisons, and Lee et al. (2025) correct the bias that a judge's imperfect sensitivity and specificity introduce into a score. Correcting a proxy's bias requires its error characteristics to carry over to the models being scored; when they are common to all of them, Proposition 3 applies and no pair can be reversed relative to another. We therefore bound instead of correcting, with homogeneity assumed only where verifiers with different extraction rules agree and carried as a parameter that we vary. The cost is fewer conclusions, each conditional on a tolerance a reader may dispute.

Partial identification and design sensitivity. The bounds of Section 4 are partial-identification statements (Manski, 2003), closest to classical results on contaminated data (Horowitz and Manski, 1995) and on binary variables misclassified at unknown rates (Molinari, 2008), with the models as sub-populations. Concurrent work applies partial identification to the prevalence of a latent attribute reported by a panel of language models, calibrated against validation labels (Chen et al., 2026); Appendix J states how our design differs.

9 Limitations

1. *The pooling assumption is not tested.* Testing it model by model needs labels drawn by model within the concordant cells, which the 400-item audit, stratified otherwise, does not provide, and the reference tolerance is a judgement, not calibrated against any external study (Section 4). If the assumption fails at that tolerance, the certified counts are too high, which is why every result is reported as a curve over ε, with a breakdown value for each ordering and the no-pooling analysis beside it.

2. *The reader verdict inherits the benchmark's reference solutions.* Every verifier and the reader verdict grade against the same reference solutions: the harness derives its gold answer from each solution's last boxed expression, and re-deriving it agrees on 1,324 of 1,324 items; a normalisation rule that damages 27 of those answers changes no audited verdict (Appendix E). An error in a reference solution therefore affects our results and the leaderboard's in the same way, and no analysis inside the pipeline can detect it. An audit of a sample of reference solutions would.

3. *The verifiers share machinery.* A response that defeats the shared symbolic library could be miscounted by all three verifiers and land in a concordant cell, which would

hide disagreement rather than create it. xFinder, which extracts differently, moved the count by one.

4. *The scope is one benchmark on one archived board.* The roster of 27 models is a convenience sample that sits low on the board and moved less than the board under the change of verifier, so its counts are unlikely to overstate the problem but do not estimate it; the secondary analysis also treats a curated benchmark as a random sample.

5. *The labels come mainly from one labeller, the author.* On overlaps of 151 and 75 items, two others agree with the reader verdicts in 89.4% and 92.0% of cases, and the labels carry no record of the time spent on each item (Appendix F).

10 Conclusion

A leaderboard names the model but not the verifier, and on this roster the choice of verifier moves the ranking without the published score showing it. A small human audit can settle part of the ranking, but only under an explicit assumption about how models differ where the verifiers agree. Without that assumption this audit certifies 0 of the 351 orderings on these problems; at the reference tolerance it certifies 174, all at once and with a finite-sample guarantee. The honest output is the curve between the two, with a breakdown value for every ordering.

Most disagreements here were responses in which a verifier found no answer. Publishing, beside each score, the verifier, its version and that count, which the pipeline computes and then discards, would let a reader bound how far a score could rise if every unread response were correct. It would not settle the disagreements that remain after extraction, which only a reader can. The assumption this paper rests on can be tested directly with labels drawn model by model in the concordant cells, and that is the natural next step.

Broader Impact Statement

Reporting which verifier produced a score, and how many responses it could not read, helps readers, model developers and leaderboard maintainers tell a ranking difference that reflects the models from one that reflects the scoring, and it costs a leaderboard almost nothing, because the pipeline already computes that count. Two harms are possible. The finding that a change of verifier reorders much of a board could be read as a reason to dismiss leaderboards altogether, although the same analysis certifies many orderings; we report what is certified alongside what is not. And knowing which verifier rewards which output format could help a developer tune a model's formatting to a verifier rather than improve its answers; publishing the verifier and its failure count makes such tuning visible rather than easier. The human labelling involved reading model responses to competition mathematics problems; no personal data was collected.

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Reproducibility

Every number in this paper is computed by released code from released result files. The code release (`https://github.com/Rajveer-code/leaderboard-orderings`) holds the code, the verifier configurations, every verifier's verdict on every response, the human labels, our analysis plan with its dated amendments, and the 400-item audit's design document.

Appendix A. Proofs

Throughout this appendix a concordant cell c is partitioned into sub-populations indexed by g, where a sub-population is a model, or a model crossed with a design stratum when the audit design was stratified more finely inside the cell. Sub-population g contains N_{gc} responses, of which a fraction p_{gc} has reader verdict $Y^* = 1$. The audit draws n_{gc} of them by simple random sampling without replacement, independently across g, and observes x_{gc} responses with $Y^* = 1$ among those drawn. We write $n_c = \sum_g n_{gc}$, $x_c = \sum_g x_{gc}$, $w_{gc} = n_{gc}/n_c$, $\hat{p}_c = x_c/n_c$ and $\tilde{p}_c = \sum_g w_{gc}\, p_{gc}$, and $\mathrm{KL}(a \,\|\, b) = a \log(a/b) + (1-a) \log\big((1-a)/(1-b)\big)$ for $a, b \in [0,1]$, with the conventions $0 \log(0/b) = 0$ and $a \log(a/0) = +\infty$ for $a > 0$. Thus $\mathrm{KL}(a \,\|\, b) = +\infty$ when $b \in \{0,1\}$ and $a \neq b$, and $\mathrm{KL}(1 \,\|\, b) = -\log b$. We assume $n_c \geq 1$; a cell without labels is given the interval $[0,1]$.

Two remarks come before the proofs. The estimand $\tilde{p}_c$ is the *design-weighted* mean of the sub-population rates, not the population mean of cell c. Those coincide when the audit samples each sub-population in proportion to its size, and they do not in general. And $\tilde{p}_c$ is a quantity the design defines; the certification result uses it only through the interval $[L_c, U_c]$, together with Assumption 1, which ties every p_{gc} to every other.

Random label counts. The setup above treats each n_{gc} as fixed. A design that samples by model and cell fixes them; the 400-item audit does not. It was stratified on the boxed verifier's verdict and the subject, so the number of its labels that land in a given sub-population of a given three-verifier cell is random. The lemma still applies, by conditioning. Take g to be a model crossed with a design stratum. Within one stratum the audit draws a uniformly random subset of fixed size, and conditional on how many of the drawn responses fall in each (sub-population, cell) domain of that stratum, the responses drawn from each domain form a uniformly random subset of that domain of the realised size, independently across domains: the uniform distribution over subsets with given domain counts factorises into a product of uniform distributions, one per domain. Strata are drawn independently. Hence, conditional on the realised vector of counts $\{n_{gc}\}$, the x_{gc} are independent hypergeometric counts exactly as the lemma assumes, and its bound holds with $\tilde{p}_c$ formed from the realised weights. Because the bound holds for every realisation of the counts, the coverage statement below holds unconditionally, for a target that is itself random through its weights. Proposition 1 is unaffected, since it needs $\tilde{p}_c$ only to be some convex combination of the sub-population rates, which it is for every realisation. The test suite checks this case directly, by simulating an audit stratified on a variable other than the cell (Appendix B).

A.1 Proof of Lemma 4

Proof of Lemma 4. Each x_{gc} is the number of responses with $Y^* = 1$ in a simple random sample without replacement of size n_{gc} from a population of N_{gc} responses of which $N_{gc}p_{gc}$ have $Y^* = 1$, so x_{gc} is hypergeometric. The argument has four steps.

Step one: sampling without replacement is dominated. Theorem 4 of Hoeffding (1963) states that if $X_1, \dots, X_n$ is a sample drawn without replacement from a finite population and $Y_1, \dots, Y_n$ is a sample of the same size drawn with replacement from the same population, then $\mathbb{E}\, f\big(\sum_i X_i\big) \le \mathbb{E}\, f\big(\sum_i Y_i\big)$ for every continuous convex f. Take $f(z) = e^{\lambda z}$, which is convex for every real λ. The with-replacement sum is binomial, so

$$\mathbb{E}\, e^{\lambda x_{gc}} \;\le\; \big(1 - p_{gc} + p_{gc}e^{\lambda}\big)^{n_{gc}}.$$

The inequality goes in the direction we need for both tails, because it holds for negative λ as well as positive.

Step two: the strata multiply. The G samples are drawn independently, so the moment generating function of x_c is the product of the individual ones: $\mathbb{E}\, e^{\lambda x_c} \le \prod_g (1 - p_{gc} + p_{gc}e^{\lambda})^{n_{gc}}$.

Step three: heterogeneous rates are dominated by their weighted mean. Let $\varphi(p) = \log\big(1 + p(e^{\lambda} - 1)\big)$, so that the product above is $\exp\big(\sum_g n_{gc}\, \varphi(p_{gc})\big)$. Writing $u = e^{\lambda} - 1$,

$$\varphi''(p) \;=\; \frac{-u^2}{(1 + pu)^2} \;\le\; 0,$$

so φ is concave on $[0, 1]$ for every λ. Jensen's inequality with the weights w_{gc}, which are nonnegative and sum to one, gives $\sum_g w_{gc}\, \varphi(p_{gc}) \le \varphi\big(\sum_g w_{gc}\, p_{gc}\big) = \varphi(\tilde{p}_c)$. Multiplying by n_c and exponentiating,

$$\mathbb{E}\, e^{\lambda x_c} \;\le\; \big(1 - \tilde{p}_c + \tilde{p}_c e^{\lambda}\big)^{n_c}.$$

The heterogeneity has been absorbed: the bound is now the moment generating function of a single binomial at the weighted mean rate.

Step four: Chernoff, including the boundary cases. Fix t with $0 < t \le 1 - \tilde{p}_c$; for larger t the event $\{\hat{p}_c \ge \tilde{p}_c + t\}$ is empty, and when $\tilde{p}_c = 1$ no t is admissible, so there is nothing to prove. For every $\lambda > 0$, Markov's inequality applied to $e^{\lambda x_c}$ gives

$$\Pr\big(\hat{p}_c \ge \tilde{p}_c + t\big) \;\le\; \exp\Big(-\lambda n_c(\tilde{p}_c + t) + n_c \log\big(1 - \tilde{p}_c + \tilde{p}_c e^{\lambda}\big)\Big),$$

so the probability is at most the infimum of the right side over $\lambda > 0$. Three cases cover every admissible t. If $\tilde{p}_c = 0$, every sub-population that carries labels has $p_{gc} = 0$, so $x_c = 0$ with certainty and the probability is $0 = e^{-n_c \mathrm{KL}(t \,\|\, 0)}$. If $\tilde{p}_c > 0$ and $\tilde{p}_c + t < 1$, the exponent is minimised at $e^{\lambda^\star} = \frac{(\tilde{p}_c + t)(1 - \tilde{p}_c)}{(1 - \tilde{p}_c - t)\,\tilde{p}_c}$, which is finite and larger than one, and substituting it gives the exponent $-n_c \mathrm{KL}(\tilde{p}_c + t \,\|\, \tilde{p}_c)$. If $\tilde{p}_c + t = 1$, the event is $\{x_c = n_c\}$ and the right side equals $\big(\tilde{p}_c + (1 - \tilde{p}_c)e^{-\lambda}\big)^{n_c}$, which decreases to $\tilde{p}_c^{\,n_c} = e^{-n_c \mathrm{KL}(1 \,\|\, \tilde{p}_c)}$ as $\lambda \to \infty$; the infimum is not attained, and no finite $\lambda^\star$ exists or is needed. This is the first inequality. For the second, with $0 < t \le \tilde{p}_c$, apply the same four steps to the complementary counts $n_{gc} - x_{gc}$, whose rates are $1 - p_{gc}$ and whose weighted mean is $1 - \tilde{p}_c$, and use the identity $\mathrm{KL}(1 - a \,\|\, 1 - b) = \mathrm{KL}(a \,\|\, b)$, which also holds under the conventions above. □

The interval, and its coverage. Inverting the lemma gives the interval used in the paper. Define

$$[L_c, U_c] \;=\; \big\{q \in [0,1] \;:\; n_c\,\mathrm{KL}(\hat{p}_c \,\|\, q) \le \log(2/\alpha)\big\},$$

which is an interval containing $\hat{p}_c$, because $q \mapsto \mathrm{KL}(\hat{p}_c \,\|\, q)$ is convex with its minimum at $q = \hat{p}_c$.

Coverage of the interval. Consider the event $\{\tilde{p}_c < L_c\}$. Since $[L_c, U_c]$ contains $\hat{p}_c$, on this event $\hat{p}_c > \tilde{p}_c$ and $n_c\mathrm{KL}(\hat{p}_c \,\|\, \tilde{p}_c) > \log(2/\alpha)$. If $\tilde{p}_c = 0$ then $x_c = 0$, so $\hat{p}_c = 0$ and the event is empty. Otherwise $a \mapsto \mathrm{KL}(a \,\|\, \tilde{p}_c)$ is continuous and strictly increasing on $[\tilde{p}_c, 1]$, from 0 to $\mathrm{KL}(1 \,\|\, \tilde{p}_c)$. If $n_c\mathrm{KL}(1 \,\|\, \tilde{p}_c) \le \log(2/\alpha)$, no $\hat{p}_c \ge \tilde{p}_c$ can exclude $\tilde{p}_c$ and the event is again empty. If not, there is a unique $t^\star \in (0, 1 - \tilde{p}_c)$ with $n_c\mathrm{KL}(\tilde{p}_c + t^\star \,\|\, \tilde{p}_c) = \log(2/\alpha)$, and the event forces $\hat{p}_c > \tilde{p}_c + t^\star$, whose probability is at most $\alpha/2$ by Lemma 4. The event $\{\tilde{p}_c > U_c\}$ is bounded by $\alpha/2$ symmetrically, and the union bound gives coverage at least $1 - \alpha$. □

Nothing in either argument requires the sub-population rates to be equal, requires sampling with replacement, or appeals to a limit. One check is exact. When no audited response in the cell has $Y^* = 1$, $\hat{p}_c = 0$ and $\mathrm{KL}(0 \,\|\, q) = -\log(1 - q)$, so the upper endpoint solves $-n_c \log(1 - U_c) = \log(2/\alpha)$, giving $U_c = 1 - (\alpha/2)^{1/n_c}$. That is exactly the Clopper–Pearson upper limit at zero successes (Clopper and Pearson, 1934), so at the boundary the design-valid interval is no wider than Clopper–Pearson; it is wider only in the interior.

A.2 Proof of Proposition 1

The proof has three parts: the region contains the truth, the optimisation separates across cells, and the cellwise optimum has a closed form. For a concordant cell with shares $a = s_{jc}$ and $b = s_{kc}$, the minimum of $ax - by$ over $R_c(\varepsilon)$ is

$$a\ell_c - b\min(h_c, \ell_c + \varepsilon) \;\text{ when } a \ge b, \qquad a\max(\ell_c, h_c - \varepsilon) - bh_c \;\text{ otherwise,} \tag{2}$$

which is what the implementation computes. A fourth part says where the region is sharp and where it is not.

The region contains the truth. Fix a concordant cell c. Model m's rate in that cell is the size-weighted mean of the rates of the sub-populations belonging to m, $p_{mc} = \sum_{g \in m}(N_{gc}/N_{mc})\,p_{gc}$, and $\tilde{p}_c$ is the design-weighted mean $\sum_g w_{gc}\,p_{gc}$ over all of them. Both are convex combinations of the same numbers $\{p_{gc}\}$. Under Assumption 1 those numbers lie in a common interval I of width at most ε, and a convex combination of points of I lies in I. Therefore p_{jc}, p_{kc} and $\tilde{p}_c$ all lie in I, which gives $|p_{jc} - p_{kc}| \le \varepsilon$ and $|p_{mc} - \tilde{p}_c| \le \varepsilon$ for $m \in \{j, k\}$. On the event $\tilde{p}_c \in [L_c, U_c]$ the second of these yields $p_{mc} \in [L_c - \varepsilon,\, U_c + \varepsilon]$, and since p_{mc} is a proportion it also lies in $[0, 1]$, so $p_{mc} \in [\ell_c, h_c]$. Together these are exactly the constraints defining $R_c(\varepsilon)$. □

Separability. By Equation (1), $\Delta_{jk} = \sum_c (s_{jc}p_{jc} - s_{kc}p_{kc})$ is linear in the unknowns and the shares are known. The constraint set is a product across cells: each concordant cell contributes $R_c(\varepsilon)$, and each discordant cell contributes the rectangle $[l_{jc}, u_{jc}] \times [l_{kc}, u_{kc}]$, which is $[0, 1]^2$ when that cell carries no labels for those models. Minimising or maximising a

sum of terms over a product of sets is done term by term. For a discordant cell the optimum is at a corner, giving $s_{jc}l_{jc} - s_{kc}u_{kc}$ and $s_{jc}u_{jc} - s_{kc}l_{kc}$. □

The closed form. Write $a = s_{jc} \geq 0$, $b = s_{kc} \geq 0$, $\ell = \ell_c$, $h = h_c$, and minimise $f(x, y) = ax - by$ over $R = \{(x, y) \in [\ell, h]^2 : |x - y| \leq \varepsilon\}$.

For fixed $x \in [\ell, h]$, the objective decreases in y, so the optimal y is the largest feasible one, $y = \min(h,\, x + \varepsilon)$. This is feasible: it is at most h by construction, at least ℓ because $x \geq \ell$ and $h \geq \ell$, and within ε of x because it is at most $x + \varepsilon$ and at least $\min(h, x) = x$. Substituting,

$$g(x) \;=\; ax - b\min(h,\, x + \varepsilon) \;=\; \begin{cases} (a - b)\, x - b\, \varepsilon, & x \leq h - \varepsilon, \\ a\, x - b\, h, & x \geq h - \varepsilon, \end{cases}$$

a piecewise linear function whose slope rises from $a - b$ to a at the kink $x = h - \varepsilon$. Since $b \geq 0$ the slope is nondecreasing, so g is convex.

If $a \geq b$, both slopes are nonnegative and g is nondecreasing, so the minimum is at $x = \ell$ and equals $a\ell - b\min(h, \ell + \varepsilon)$.

If $a < b$, the first slope is negative and the second is nonnegative, so the unconstrained minimiser is the kink, and the constrained minimiser is $x^\star = \max(\ell,\, h - \varepsilon)$. When $h - \varepsilon \geq \ell$ the value is $a(h - \varepsilon) - bh$. When $h - \varepsilon < \ell$ the minimiser is ℓ, and since then $\ell + \varepsilon > h$ the value is $a\ell - bh$. Both cases are written by the single expression $a\max(\ell,\, h - \varepsilon) - b\, h$.

The maximum follows by symmetry. R is invariant under exchanging its two coordinates, and $\max_R(ax - by) = -\min_R(bx - ay)$ after that exchange, so the maximum is obtained by applying the formula above with a and b swapped and negating the result. □

Sharp at $\varepsilon = 0$, conservative above it. At $\varepsilon = 0$ we have $\ell_c = L_c$ and $h_c = U_c$, and $R_c(0)$ is the diagonal segment $\{(q, q) : q \in [L_c, U_c]\}$. Every point of it is attained in the relaxation where each p_{gc} may take any value in $[0, 1]$: set every sub-population rate in the cell to q, which satisfies $H(0)$ and makes $p_{jc} = p_{kc} = \tilde{p}_c = q$. The interval for Δ_{jk} is therefore attained and cannot be narrowed, which is what sharpness means here. The qualification about the relaxation matters, because each p_{gc} is really a fraction over finitely many responses and only finitely many values of q are exactly attainable. This is the same granularity that Section 4 uses to argue that $H(0)$ is an idealisation rather than a hypothesis about the world.

For $\varepsilon > 0$ the region is an outer bound, and the reason is that it forgets the weights. $R_c(\varepsilon)$ imposes only that p_{jc} and p_{kc} lie within ε of each other inside $[\ell_c, h_c]$. The actual constraint is stronger: the same sub-population rates that produce p_{jc} and p_{kc} must also produce a design-weighted mean $\tilde{p}_c$ lying in $[L_c, U_c]$. When one model supplies most of the audit weight in the cell, its own rate is pinned close to $\tilde{p}_c$ and cannot travel the full distance ε away from $[L_c, U_c]$. The computed range therefore contains the sharp identified range. Every consequence runs in the safe direction: reported ranges are at least as wide as the truth requires, fewer orderings are certified than a sharp analysis would certify, and each breakdown value in Definition 2 is a lower bound on the sharp one. □

A.3 Proof of Proposition 3

Proof of Proposition 3. Let a_m be the fraction of model m's responses the single verifier credits. Conditioning on the verifier's verdict and using the assumed common predictive

values,

$$A_m \;=\; \Pr(Y^* = 1 \mid v = 1)\, a_m + \Pr(Y^* = 1 \mid v = 0)\,(1 - a_m) \;=\; \pi_0 + a_m(\pi_1 - \pi_0),$$

so $\Delta_{jk} = A_j - A_k = (\pi_1 - \pi_0)(a_j - a_k)$. □

The same statement in the other natural parameterisation makes the consequence easier to see. Suppose instead that the verifier has common error rates, $q = \Pr(v = 1 \mid Y^* = 0)$ and $r = \Pr(v = 0 \mid Y^* = 1)$, shared by all models. Then $a_m = (1 - r)A_m + q(1 - A_m) = q + (1 - q - r)A_m$, so $a_j - a_k = (1 - q - r)\,\Delta_{jk}$. The observed gap is the gap under the reader verdict, scaled by one number.

Either way, a single multiplicative constant relates every observed pairwise gap to the corresponding gap under the reader verdict. Three consequences follow, and they are the reason the design uses more than one verifier.

The sign of every pairwise comparison is determined by the sign of that constant. When $\pi_1 > \pi_0$, equivalently $q + r < 1$, every ordering the verifier reports is the ordering under the reader verdict. When $\pi_1 < \pi_0$ every ordering is reversed at once. A model that a common mechanism treats unusually badly does not exist inside this parameterisation.

Pair-specific reversals are therefore outside the model. A reversal that affects one pair and leaves another intact cannot be produced by any choice of π_0 and π_1, so observing one is evidence that the mechanism differs across models. Section 6 reports such reversals.

At $\pi_1 = \pi_0$, equivalently $q + r = 1$, every Δ_{jk} is zero and the verifier carries no information about the ordering at all.

The role this plays in the design is the following. Pooling under a single verifier is Assumption 1 at $\varepsilon = 0$ with $K = 1$, where the pooled cells cover the entire population and nothing is left model-specific. That assumption does not merely simplify the analysis; by the proposition it removes the phenomenon the paper is about. With K verifiers the pooling is confined to the cells where they agree, and the cells where they disagree, which is where the disputed responses live, stay model-specific by construction.

A.4 Proof of Proposition 5

The audit half, in finite samples. List the parameters a certification uses: the pooled rate $\tilde{p}_c$ for each concordant cell, which is shared by every pair on the roster, and the per-model rate p_{mc} for each audited discordant cell. Distinct parameters are estimated from disjoint sets of labels, and conditional on the realised number of labels in every domain those sets are independent (the paragraph on random label counts above), so the corresponding intervals are conditionally independent; the argument below is conditional on those counts, and it holds for every realisation of them. Give each of the P parameters an interval at level $1 - \alpha'$ with $\alpha' = 1 - (1 - \alpha_{\text{audit}})^{1/P}$, which is the allocation of Šidák (1967) and is exact rather than conservative under independence. By Lemma 4 each interval covers its parameter with probability at least $1 - \alpha'$, and by independence all of them cover simultaneously with probability at least $(1 - \alpha')^P = 1 - \alpha_{\text{audit}}$.

On that event, the hypothesis of Proposition 1 holds for every cell at once, so the identified range computed for every pair contains the corresponding Δ_{jk} simultaneously. Any ordering whose range lies strictly above zero is therefore correct. No step used a limit, so the statement is exact in finite samples. □

The item-sampling half, asymptotically. Let δ_{jk} denote the pairwise difference over the distribution from which the benchmark's items were drawn, and Δ_{jk} the difference on the realised items. With D_i the difference of the two models' reader verdicts on item i, the sample mean of D is Δ_{jk} and its expectation is δ_{jk}. The central limit theorem gives $\Pr(|\delta_{jk} - \Delta_{jk}| > z_{1-\alpha_s/2}\sqrt{\mathrm{Var}(D)/N}) \to \alpha_s$, and replacing $\mathrm{Var}(D)$ by the upper bound of Section 4.5 can only widen the interval. Combining the two halves by a union bound,

$$\Pr\big(\delta_{jk} \notin [\mathrm{LB}_{jk} - h_{jk},\, \mathrm{UB}_{jk} + h_{jk}]\big) \;\le\; \Pr\big(\Delta_{jk} \notin [\mathrm{LB}_{jk}, \mathrm{UB}_{jk}]\big) + \Pr\big(|\delta_{jk} - \Delta_{jk}| > h_{jk}\big),$$

which is at most $\alpha_{\text{audit}} + \alpha_{\text{sample}}$. The two events concern different sources of error, one the labels and the other the items, so nothing is counted twice; adding the levels rather than convolving the two error distributions is what makes the combination conservative. □

For a family of orderings, the audit half is already simultaneous because the concordant intervals are shared and the discordant allocation runs over all audited per-model parameters. The sampling half is made simultaneous by a Bonferroni division of α_{sample} across the family. Rank intervals inherit the family level directly: on the event that every certified ordering is correct, at least one model is ranked above m for each ordering $k \succ m$ that was certified and at least one below for each $m \succ k$, so m's true rank lies between one plus the first count and the number of models less the second.

A.5 The paired variance bound

Section 4.5 bounds the item-sampling variance without observing the reader verdict on unaudited items. Two elementary steps do the work.

First, D_i takes values in $\{-1, 0, 1\}$, so D_i^2 is the indicator that the two models are scored differently and $\mathrm{Var}(D) \le \mathbb{E}[D^2] = \Pr(D \neq 0)$.

Second, for any verifier v and any response pair,

$$\mathbf{1}[Y_j^* \neq Y_k^*] \;\le\; \mathbf{1}[v_j \neq v_k] + \mathbf{1}[Y_j^* \neq v_j] + \mathbf{1}[Y_k^* \neq v_k].$$

If the left side is one and all three terms on the right are zero, then $v_j = v_k$, $Y_j^* = v_j$ and $Y_k^* = v_k$, which forces $Y_j^* = Y_k^*$ and contradicts the left side. Averaging over items, the disagreement rate between the two models under the reader verdict is at most the verifier's own disagreement rate between them, which is observed on every item, plus each model's disagreement rate with that verifier, which the identified bounds of Proposition 1 bound above at the same ε. Taking the smallest bound over the available verifiers, and capping at one, gives the quantity used in the half-width. It is paired, because the verifier disagreement term is computed item by item rather than from the two models' marginal accuracies, and it is conservative, because every ingredient is an upper bound.

Appendix B. Numerical checks of the stated results

Appendix A proves the results. This appendix describes what is checked numerically, which is a different question with three parts: whether the code computes what the propositions say, whether the prose says what the code computes, and whether a change to the data pipeline silently moves a number that was fixed earlier. The released test suite holds 917 cases in 102 test functions.

B.1 The propositions against independent computation

The first group of tests, 618 cases, compares the implementation with computations that do not share its code path.

The closed form of Proposition 1 is checked against brute force on randomly drawn cells: a dense grid over $R_c(\varepsilon)$, together with the region's exact polygon vertices so that an optimum sitting between grid points cannot be missed. A separate check asserts where the minimiser sits, not only what the minimum equals, because a formula can return the right value from the wrong argument and stay wrong at the next input. Three degenerate configurations are pinned explicitly: the cell where no verifier credits, in which $L_c = 0$ and the clipping binds for every positive ε; a cell in which both models have zero share and must contribute nothing; and an ε large enough that the region is the whole unit square.

Monotonicity of the lower bound in ε is checked over a grid, since Definition 2 is well posed only if it holds. The breakdown value itself is checked by running the bisection and, separately, scanning a fine grid for the first ε at which the margin fails; the two must agree to a stated tolerance. The interior slope of a concordant cell's contribution is checked to be $-\max(s_{jc}, s_{kc})$, which rules out the valid but looser slope $-(s_{jc} + s_{kc})$; the looser slope would pass every other check.

Proposition 3 is checked in both parameterisations over random draws, for every pair in a synthetic board, which is the form in which the claim is actually used: not that some pair is scaled, but that all of them are scaled by the same number. The claim that sampling without replacement can only help is checked against exact binomial coefficients rather than against the bound that replaces them.

Lemma 4 gets four checks. At zero and at full successes the interval must equal the Clopper–Pearson limit exactly, which is the boundary identity derived in Appendix A. The interval must contain the point estimate and must widen with the level. A Monte Carlo draws pooled hypergeometric samples from sub-populations with deliberately different rates and requires the realised miss rate not to exceed nominal; that check exercises both departures the lemma is about at once, and it is the one that would fail if the Jensen step were wrong. A second Monte Carlo stratifies the draw on a variable other than the cell, so that the number of labels in each domain is random, and requires the interval to cover the mean formed with the realised weights, which is the conditioning argument of Appendix A. A last check compares both tail bounds with exact probabilities at the boundaries treated in step four ($\tilde{p}_c = 0$, $t = 1 - \tilde{p}_c$ and $t = \tilde{p}_c$), and confirms coverage in the case where no $t^\star$ exists; doubling the exponent of the bound makes it fail.

The level allocation is checked separately: the number of parameters a statement relies on is computed from one function for a single ordering and for the whole roster, with and without pooling, and a test confirms that an allocation over one pair's parameters falls short of the stated family-wise level once per-model discordant intervals exist.

Rank intervals are checked against a small board whose certified set and correct answer are written out by hand.

B.2 The printed formulas against the implementation

Transcription is where errors enter a paper, so a second group of tests, 203 cases, re-implements the formulas exactly as printed here, without importing the library, and compares

them with the implementation on random inputs. It covers the closed form (2), Proposition 3, the claim that $\varepsilon = 0$ collapses to the pooled analysis, and the claim that certification is monotone. A formula changed in the paper but not in the code, or the reverse, makes these tests fail.

B.3 The regression anchor

A third group, 7 cases, rebuilds the earlier seventeen-model, two-verifier configuration from the *current* verdicts and audit map and requires it to reproduce the counts that the earlier two-verifier pipeline produced under its string semantics: 84 orderings for the secondary target and 101 for the primary target, one ordering at a time. Its purpose is to catch any change to the verdicts, the audit map or the verifiers that silently alters a previously reported result, which is the class of failure the defect of Section 2 belongs to.

B.4 What these checks do not establish

They do not prove the propositions; Appendix A does. They do not validate the empirical results, which come from the released result files. And a check written from the same misunderstanding as the code will pass. That is the reason for the division of labour above: the propositions are proved on paper, the second group re-implements from the printed text rather than from the library, and the anchor compares against numbers fixed before the current code existed.

Appendix C. Inference details

C.1 How the levels are allocated

The audit and the item sample are controlled separately, at $\alpha_{\text{audit}} = 0.025$ and $\alpha_{\text{sample}} = 0.025$, and a claim that generalises beyond the benchmark's items is made at one minus their sum. Proposition 5 is what licenses adding them.

Within the audit, the level is divided across the parameters a claim uses, by the allocation $\alpha' = 1 - (1 - \alpha_{\text{audit}})^{1/P}$. For a single ordering those parameters are the pooled interval of each concordant cell, plus each of the two models' per-model discordant intervals. For a statement about every ordering at once they are every parameter on the roster: the pooled intervals, which every pair shares, plus every per-model discordant interval, each of which enters every ordering its model appears in. Both kinds of statement count parameters the same way, and a test checks that a per-pair count would fall short of the stated family-wise level.

In the configuration reported here no discordant cell contributes a parameter, because the 400-item audit was not drawn by model and cell; its labels in a discordant cell are described in Appendix F and do not enter the certification. P is therefore the number of concordant cells for both kinds of statement, every pooled interval is constructed at level 0.9874, and the audit side of a family-wise statement is simultaneous without further widening. The family-wise setting therefore changes only the sampling term, which takes a Bonferroni division of α_{sample} across the 351 orderings. An audit that supplied per-model discordant labels would add parameters, and the audit intervals of a family-wise statement would widen with them.

Without pooling, each model's concordant intervals enter as model-specific parameters, so a statement about every ordering at once allocates over two per model on the roster rather than four per pair. The variance bound in the sampling term is always computed from the same intervals as the range it widens.

Rank intervals are read off the certified set and inherit whichever level that set was constructed at, since they are a deterministic function of it.

C.2 The item-sampling term, for the secondary target

Under the item-sampling interpretation, a claim about which model is better in general carries a second source of uncertainty that the audit cannot reduce. We bound it by a paired argument. For a pair of models, let D_i be the difference of their reader verdicts on item i. Then $\mathrm{Var}(D) \le \Pr(D \neq 0)$, and $\Pr(D \neq 0)$ is bounded above by the observed rate at which a verifier's verdicts differ between the two models, plus each model's maximum possible disagreement with that verifier, both of which the identified range already supplies (Appendix A.5). The resulting half-width is paired, so it exploits the fact that both models answer the same items, and it is conservative, because it uses upper bounds rather than estimates. It rests on a central limit approximation, so the combined interval is asymptotic, not exact.

The two sources of error are combined by a union bound, and they are distinct. The audit interval concerns the unknown correctness rates given these responses; the item-sampling term concerns generalisation beyond these items. Nothing is counted twice. The two halves of Proposition 5 are accordingly different in kind: the first is a finite-sample guarantee that follows from Lemma 4, and the second inherits the central limit approximation of the sampling term. Rank intervals are reported as the family-level object, because a reader looking at a leaderboard is reading all the orderings at once, not one.

C.3 A sampling term that does not rest on a limit

The paired half-width above is conservative in its variance but asymptotic in its distribution, and a reader who declines the central limit approximation needs a number, not a caveat. Because the paired difference D_i lies in $[-1, 1]$, Hoeffding's inequality bounds the deviation of its mean without any variance estimate at all, giving the half-width

$$h \;=\; \sqrt{\frac{2\log(2/\alpha_{\text{sample}})}{N}},$$

which is finite-sample, distribution-free, and the same for every pair. It is also much wider, and the cost is visible: for the secondary target, one ordering at a time at $\varepsilon = 0$, substituting it certifies 100 orderings where the paired asymptotic term certifies 153. Both are reported: the distribution-free count stands without the approximation, and the gap between the two is what the approximation buys. The primary target needs neither, because it carries no sampling term.

The pairing matters separately from the distributional question. An unpaired term, built from each model's marginal accuracy under one verifier, is kept only to reproduce the regression anchor of Appendix B; it certifies 160 orderings. It is not used, because two

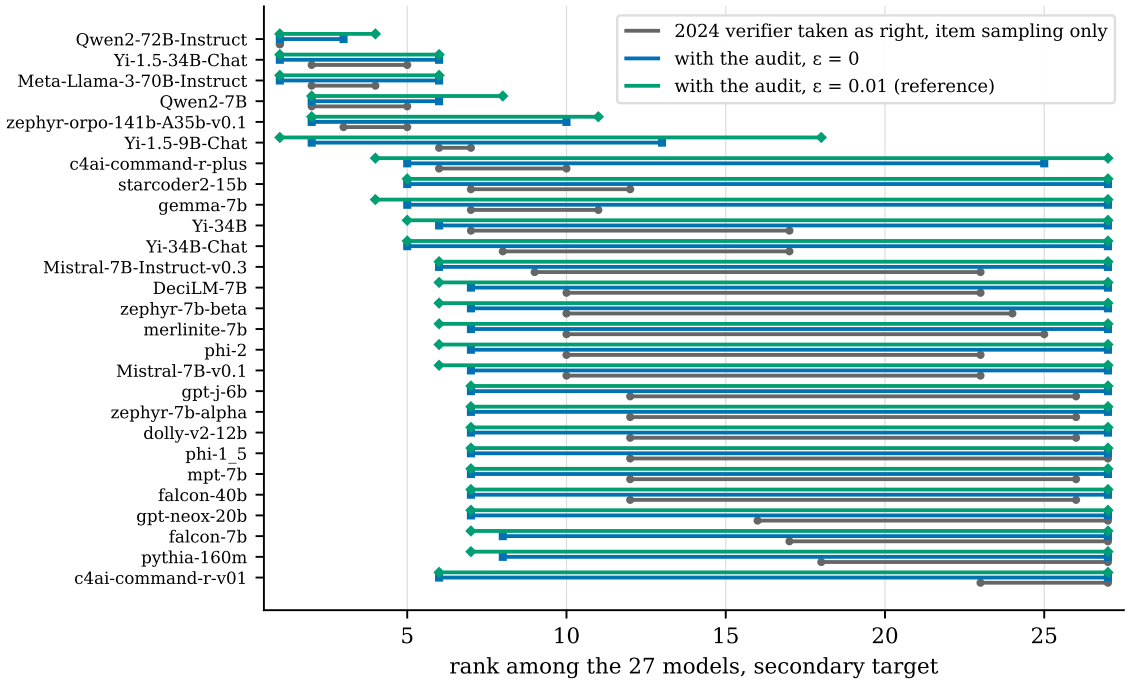


Figure 3: Rank intervals for the secondary target, which also allows for the items being a sample of possible questions, holding for all orderings at once at an asymptotic level. Grey takes the 2024 verifier as correct and allows only for item sampling. Without pooling across models (not drawn), every model's interval spans all 27 ranks. Compare Figure 2, which is the primary target on these items.

models answering the same items differ on far fewer of them than their marginal accuracies suggest, and a term that ignores the pairing throws that away.

C.4 What the coverage simulation does

The simulation runs 9 scenarios at 2,000 replicates each, at the full scale of the roster. It checks whether the guarantee is attained and how it degrades; it is not a proof.

The frame is the full set of real responses, with the real cell structure and the real audit allocation, so that the strata, their sizes and their sampling rates are the ones the paper actually uses. What must be invented is the reader verdict on unaudited responses, and it is invented in two deliberately different ways.

In the first, each concordant sub-population is given a number of errors that makes the realised error fractions span an interval of width exactly $\varepsilon_{\text{true}}$, so that Assumption 1 holds exactly as a statement about finite populations. This is the validity check: the procedure is analysed at the same ε, and no certified claim should be wrong.

In the second, concordant errors are drawn independently with a small propensity that is the same for every model. No model differs from any other in expectation, which is the

most favourable situation pooling could hope for, and yet the realised fractions differ by chance, so $H(0)$ fails as a statement about the finite populations that actually exist. This is a diagnostic rather than a validity check, and it is the one that produced the paper's reason for not reporting $\varepsilon = 0$ as a result.

Discordant cells are given independent per-model rates drawn uniformly, which is the worst case the analysis is built for and the reason those cells enter at $[0, 1]$ when unaudited. Items are resampled from the frame with replacement and paired across models, so the item-sampling term has something real to cover. An audit-only claim is scored against the difference on the realised items; a combined claim is scored against the superpopulation expectation. Confusing those two would make the procedure look better than it is, which is why they are kept apart in the output.

One number bounds what the simulation can resolve. At the nominal level and this many replicates, the Monte Carlo standard error of a family-wise error rate is 0.0035, so a reported rate of zero means below resolution rather than exactly zero, and the scenarios are compared against nominal with that in mind. Zero failures in 2,000 replicates is likewise not a zero error probability: it bounds the family-wise error rate below 0.0015 at 95% confidence, one-sided, so the procedure is consistent with its stated level and conservative rather than tight.

The second construction is the one that separates the two configurations reported in Section 4.5. With equal underlying rates analysed at $\varepsilon = 0$, the two-verifier analysis exceeds its nominal level and the three-verifier analysis does not, so exact homogeneity is safe in one configuration and unsafe in the other. Equal underlying rates do not imply equal realised fractions, so $H(0)$ can fail in the world most favourable to pooling, which is why the paper does not lead with it.

Appendix D. Sensitivity analyses

Each analysis in this appendix re-runs the whole pipeline with a single element replaced rather than recomputing the headline numbers a different way. All but the last row of Table 3 are sensitivity analyses named in our analysis plan; the others it names are the two-verifier analysis (Section 5) and the alternative sampling terms (Appendix C). The table reports them, with the pooled audit counts each configuration produces alongside the orderings it certifies, because the counts are what the certification is downstream of and reporting only the certified number would hide where a movement came from.

Four features of the table matter.

The primary result is stable. On these items at the reference tolerance, every alternative that keeps the verifiers fixed lands within a few orderings of 174. The choice of interval makes no difference there at all: Clopper–Pearson, which is tighter but exact only under assumptions this design does not satisfy, also certifies 174. The headline does not depend on having used the conservative interval, so the conservative one is used because it is the valid one here, not because the result needs it.

The secondary result moves more, and mostly for one reason. Judging the reader's transcription with Math-Verify costs the primary target a handful of orderings and the secondary target many more; Appendix E explains why the same labels matter so differently to the two.

Table 3: Sensitivity analyses. Each row re-runs the whole pipeline with one element of the analysis replaced, and reports the pooled audit counts it produces in the two concordant cells together with the certified orderings, first for the primary target (all orderings at once) and then for the secondary target (one ordering at a time). [a]The pooled interval, in place of Lemma 4. [b]The routine that decides Y^*. [c]NONE and CONTRADICTORY excluded rather than scored incorrect. [d]Changes the boxed verifier *and* the equivalence routine, so cells move too. [e]A different set of models: its counts are out of a smaller family and are not comparable down the column.

			primary target		secondary target	
analysis	111	000	$\varepsilon = 0$	$\varepsilon = 0.01$	$\varepsilon = 0$	$\varepsilon = 0.01$
primary analysis	142/144	0/223	208	174	153	131
Clopper–Pearson interval[a]	142/144	0/223	209	174	155	131
Math-Verify equivalence[b]	141/144	1/223	206	172	143	128
indeterminates dropped[c]	142/144	0/186	207	174	152	130
string semantics[d]	141/142	0/223	196	168	141	127
seventeen-model subset[e]	142/143	0/145	99	89	76	68

The string-semantics row moves both targets furthest, but it changes a verifier as well as the equivalence routine, so it is not a clean test of either.

The last row is a different question from the others. It restricts the roster to a seventeen-model subset, so its counts are out of 136 orderings rather than 351 and cannot be compared down the column. It is in the table because it is the configuration the regression anchor of Appendix B reproduces.

What none of these rows do is vary Assumption 1. They vary the analysis applied to a fixed assumption. The assumption itself is varied by the ε curve of Figure 1 and summarised per ordering by the breakdown value, and that is the sensitivity analysis the paper is actually built around. The rows here answer the narrower question of whether a defensible alternative choice inside the pipeline would have produced a different story, and none of them does.

Appendix E. Sensitivity to the equivalence routine and to indeterminate answers

Section 2 defines the reader verdict in two steps, and only the first is human. A labeller transcribes the final answer a response commits to, and an equivalence routine then compares that transcription with the benchmark's gold answer; Y^* is whatever the routine returns. The routine is a choice, and since this paper is about scoring choices that go unreported, ours is reported here.

The primary analysis uses the benchmark harness's own routine, the same symbolic comparison the leaderboard applies to its gold answers. We use it because the question is whether a response would have been credited had a reader, rather than an extraction rule, decided what it says, and holding the equivalence step at the benchmark's own routine isolates the extraction step, which is the thing under study. It is not necessarily the best routine available.

A different equivalence routine. Deciding Y^* with Math-Verify instead credits 1 audited response in the cell where no verifier credits, against 0 under the harness routine, and 141 of 144 in the cell where all three credit, against 142. Both differences push the bounds the same way. On these items the certified count moves from 208 to 206 at $\varepsilon = 0$, and from 174 to 172 at the reference tolerance. For the secondary target it moves from 153 to 143 at $\varepsilon = 0$, one ordering at a time.

The same two labels matter very differently to the two targets, and the reason shows where the pipeline is fragile. The cell where no verifier credits holds 32,685 responses, and one audited response with reader verdict 1 raises the upper end of its interval substantially. In the identified range of a pairwise difference, that interval enters multiplied by the *difference* between the two models' shares of the cell, which is usually small, because most of every model's responses land there; so the primary target barely moves. In the sampling term of the secondary target, the same interval enters each model's own bound on its disagreement with a verifier, multiplied by that model's *own* share, which is close to one, and the two models' bounds add. That is where most of the secondary target's movement comes from. The equivalence routine is therefore a minor choice for the primary claim and a consequential one for the secondary claim.

String comparison. The third row of the same family is reported in Table 3 and should be read with care, because it is not a one-factor change. The boxed verifier and the equivalence routine share a normalisation and comparison implementation, so switching to string comparison changes which cell a response falls into as well as how its transcription is judged. The label counts move accordingly: the cell where all three verifiers credit holds 142 audited responses rather than 144. On these items at the reference tolerance certification falls to 168; for the secondary target at $\varepsilon = 0$ it falls to 141. We report the row because string comparison is what the defective verifier of Section 2 actually computed, and we do not attribute the movement to the equivalence routine alone.

Responses with no answer, and responses with two. A labeller who finds no final answer records NONE, and one who finds two answers both presented as final records CONTRADICTORY. The primary analysis counts both as incorrect. The alternative is to drop them, which reduces the audited labels in the cell where no verifier credits from 223 to 186. On these items at the reference tolerance the certified count stays at 174; for the secondary target at $\varepsilon = 0$ it moves from 153 to 152.

The two agree closely for a mechanical reason. Dropping the indeterminates removes labels without removing any credited response, so the numerator stays at its floor while the denominator shrinks; the interval widens slightly and nothing else moves. The choice would matter more in a cell where indeterminate responses were a large share of the credited ones, and the design does not guarantee that they never are.

The reason the primary analysis counts them as incorrect is not statistical convenience. Every verifier in the pipeline scores a response with no extractable final answer as zero, and the comparison the paper is making is against what the leaderboard reports. Dropping such responses would compare against a quantity no leaderboard computes, so Section 2 makes this choice part of the definition of the reader verdict.

A normalisation rule that edits gold answers. The harness normaliser deletes every occurrence of the letters "ft" (feet), including inside LaTeX commands, so `\infty` becomes

\iny and \left becomes \le, and the harness applied it to the gold answers when it built the dataset; a public discussion of the leaderboard's archive reports the resulting damage (svk-1110, 2026). Normalising each reference solution's boxed answer with and without the rule, we find 27 affected items, the number the discussion reports, and on each the stored answer is the damaged one. The rule is also applied to every transcription, so the reader verdict compares like with like, but we checked rather than assumed: 6 audited labels fall on these items, 1 of them in a concordant cell, and recomputing their reader verdicts without the rule changes 0 of them. Every certified count is therefore unchanged.

What none of this settles. Every row above holds the benchmark's gold answers fixed. An error in a gold answer propagates identically through the primary analysis and through all of its alternatives, and no sensitivity analysis run inside the pipeline can detect it. It is the largest thing this design does not check (Section 9).

Appendix F. Data, verifiers, and the human audit in detail

How the roster came about. By the maintainers' count, the February 2025 re-scoring covered all 3,751 models submitted to the leaderboard up to then (Kydlíček et al., 2025). An earlier study catalogued all 3,996 of the leaderboard's per-model result repositories, ranked them by download count, and kept the top 60 subject to at most 4 per organisation, under a rule written down before any re-scoring; the cap spread the selection over 28 organisations. A download sweep over those models was interrupted, and the 28 whose downloads completed were scored. The 400-item audit drew its items from those, and the roster is the 27 that received at least one audited item. Every roster model lies within the first 40 of the download ranking. Against the leaderboard's final public table, dated 20 March 2025, which lists 4,497 models with a MATH-Hard score because submissions continued after the re-scoring, 16 of the 27 fall in the bottom quartile of displayed accuracy and 4 in the top. The record behind each count is released with its checksums. A prespecified rule covered missing responses before the data were assembled, and it did not fire.

The verifiers' provenance. The maintainers described the 2024 verifier as unable to extract answers written in boxed notation and reported that the re-scoring reordered the top of the board (Kydlíček et al., 2025). From the re-scoring on, the leaderboard displayed scores from one verifier and published item-level verdicts from another. Table 4 gives the version, a checksum and the check behind every source the analysis rests on, including the re-execution that reproduces all 35,748 published verdicts.

The fourth verifier and its sixteen cells. xFinder reads each response and returns an answer or an abstention. An extraction that is empty or reads "no valid answer" (ignoring case, brackets and parentheses) is an abstention and receives verdict 0, as does an extraction whose comparison with the gold answer is stopped after sixty seconds; every other extraction is judged by the equivalence routine that decides the reader verdict. Appending xFinder's verdict to the three-verifier string gives sixteen cells, of which 15 are occupied. The concordant cells are 0000 and 1111, and Assumption 1 applies to them; the other fourteen are discordant and enter each model's range at $[0, 1]$, as the three-verifier discordant cells do. Every audited response in a three-verifier concordant cell stays concordant (144 labels in 1111, 223 in 0000), so the pooled intervals are unchanged, and the four-verifier count moves only because 31

Table 4: Provenance of every source the analysis rests on. Hashes are the first twelve hexadecimal digits of the sha256 of the file in the release, with line endings normalised; a dash marks an upstream artefact identified by its pinned version instead. Verifier environment: sympy 1.13.1, latex2sympy2_extended 1.11.0.

artefact	version	hash	check
published responses	27 per-model result repositories, 189 subject files; revision of each file recorded in the release	--	sample files unmodified since 2024 in 27 of 27 models
2024 verifier	lm-evaluation-harness `3c8db1bb7b`	`59604119822c`	re-execution reproduces 35,748 of 35,748 published verdicts
Math-Verify	math-verify 0.9.0 via lm-evaluation-harness `61ee867816`	--	reproduces the re-scored aggregate in 168 of 175 model-subject combinations
boxed verifier	lm-evaluation-harness helpers, in the released code	`ac7865eec34c`	brute-force and printed-formula tests (Appendix B)
xFinder	`IAAR-Shanghai/xFinder-qwen1505`	--	extractions judged by the equivalence routine
scores before rescoring	`open-llm-leaderboard/results` at `c6b22601e3`	--	last revision before February 2025
scores after rescoring	`open-llm-leaderboard/contents` at `9c09a7cae4`	--	final revision of the public table
all verifier verdicts	released verdict file	`878751536bfd`	regenerated by the released pipeline

Table 5: What the three verifiers share and where they differ. All three rest on the same symbolic algebra library, and the 2024 and boxed verifiers also share the harness's normalisation and the dataset's gold-answer field. Their agreement is informative because their extraction rules and parsers differ, not because they are independent.

step	2024 verifier	boxed verifier	Math-Verify
answer extraction	sentence pattern	last boxed expression	LaTeX and expression parse
normalisation	harness routine	harness routine	latex2sympy2
equivalence	sympy	string, then sympy	Math-Verify
gold answer	dataset field	dataset field	parsed from the solution

responses leave the concordant cells where xFinder disagrees with the other three. The primary arm uses the system prompt the extractor was fine-tuned with; the other uses its shipped default.

The defect in our own verifier. While tracing where each verdict came from, we found that the boxed verifier had been silently degraded. Its symbolic comparison guards itself with a timer that exists only on Unix; on Windows the guard raised an exception, a broad error handler swallowed it, and every comparison returned "not equivalent", so every verdict

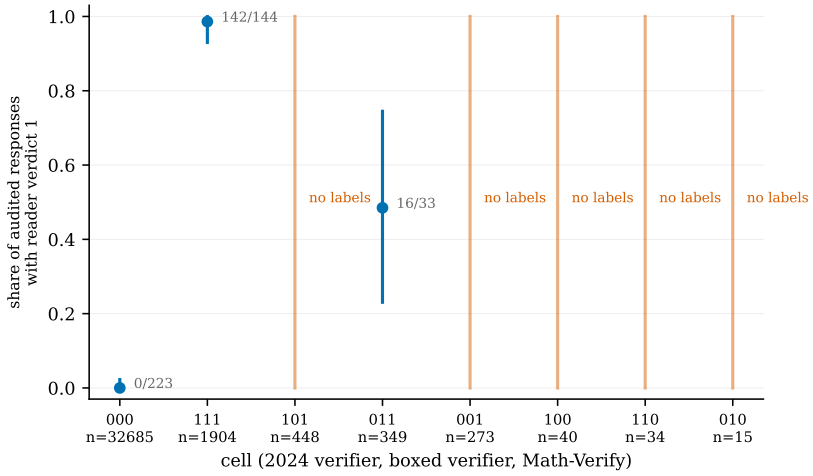


Figure 4: Share of audited responses in each cell with reader verdict 1, that is, whose transcribed answer the benchmark's equivalence routine accepts. Intervals are from Lemma 4. Five cells carry no labels in the 400-item audit and are bounded only by $[0, 1]$; the labelled discordant cell is shown for description and does not enter the certification.

was in effect a string comparison. Correcting it changes the number of responses on which it disagrees with the 2024 verifier, on a seventeen-model subset kept as a regression anchor, from 817 to 704. All results use the corrected verifier. Correcting it also changes the secondary-target certified count on the seventeen-model regression anchor (Appendix B), one ordering at a time, from 84 to 89. The anchor is the only configuration in which both versions can be computed on the same labels.

The 400-item audit's discordant labels. Some of the 400-item audit's labels land in a discordant cell: 33 of them, all in one cell, spread over 11 models at a median of 2 per model. A discordant cell enters the identified range only through a per-model interval, and the analysis takes those intervals only from labels drawn by model and cell, which these are not. They therefore appear in Table 1 and Figure 4 but not in the certified counts.

How reliable the 400-item audit's labels are. The author labelled all 400 items. A second labeller labelled an overlap of 151 of them and a third a nested overlap of 75, with both overlaps drawn before any label was seen; each was shown the problem and the response and nothing else. Scoring the transcribed answers with the equivalence routine, the first and second labellers agree on 77.5% of the overlap (Cohen's $\kappa = 0.61$), the first and third on 89.3% ($\kappa = 0.82$), and the second and third on 88.0% ($\kappa = 0.79$). What enters the certification is the reader verdict Y^*, and on it the second labeller agrees with the first on 135 of 151 items ($\kappa = 0.77$) and the third on 69 of 75 ($\kappa = 0.83$). The disagreements are not in the cells that carry the result. In the cell where no verifier credits there are none (85 and 42 overlapping labels), and in the cell where all three credit there are 4 of 50 and 0 of 24; the others fall in discordant cells, which do not enter the counts. Replacing the

first labeller's labels on the overlap by the second labeller's moves the certified count at the reference tolerance from 174 to 172 (from 208 to 205 at $\varepsilon = 0$), and replacing them by the third's leaves it at 174. The overlaps are subsets of the 400 items, so this bounds the sensitivity to the labeller on those subsets only. The audit was labelled in a spreadsheet, so its labels carry no record of the time spent on each item.

What was fixed, and when. Table 6 lists each analysis choice with the document that fixed it and whether the 400-item audit's labels already existed at the time. Only the audit's design came before its labels. This study's analysis plan, written for the larger audit before any of that audit's labels existed, came after the 400-item audit's labels, and so did everything it fixes: the method of this paper, the tolerance and the sensitivity analyses. The choice of primary target came later still, after the audit's results for both targets were known. The analysis reported here should therefore be read as designed with those labels in hand. What protects it from tuning is that every count is reported as a curve over ε, beside the analysis without pooling, and both targets in full. The code release holds both documents with their dates.

Table 6: When each analysis choice was fixed, relative to the 400-item audit's labels.

choice	fixed in	the audit's labels
design of the 400-item audit	design document of the earlier study	not yet collected
the method: cells, pooled intervals, level allocation	this study's analysis plan and its first four amendments	already existed
tolerance $\varepsilon = 0.01$ and the sensitivity analyses	fourth amendment	already existed
fixed-item target as primary	sixth amendment	existed, with results for both targets known
larger audit by model and cell, and the discordant census	fourth and sixth amendments	not labelled; no result uses them

Appendix G. The roster against the board

Figure 5 places the roster in the distribution of displayed MATH-Hard accuracy over every model in the leaderboard's final public table (20 March 2025), and Figure 6 plots each traced model's score under the 2024 verifier against its score under Math-Verify, both read from the same results file. Both use the leaderboard's own records at fixed revisions; neither involves re-scoring by us. The roster sits low on the board, and it moved less under the change of verifier than the board did.

Tracing each displayed score. The re-scoring rewrote results files in each model's details repository on 13 February 2025, changing the MATH-Hard scores and their task version; in every roster file we could compare, the run's configuration was left unchanged (Table 4). For each of the 3,413 models with a MATH-Hard score both in the results revision of 31 January 2025 and in the final public table, we listed the results files in its details repository with the date of the last change to each, read every file changed on the re-scoring day, and

compared its MATH-Hard score with the displayed one, taking the largest when a model is listed at more than one numerical precision. The displayed score equals the score of a re-scored file for 3,393 models. For 8 models no file was re-scored, for 5 the displayed score matches none of the re-scored files, and for 7 the repository could not be read. Over every displayed row of the models we could read, 76 rows match no re-scored file, consistent in kind with a public audit that reports displayed rows left on the old verifier (svk-1110, 2026).

The reversal statistic. For 3,366 of the traced models we could also read the re-scored file as it stood before the re-scoring. Let a_m be model m's MATH-Hard score in that file before the re-scoring and b_m its score in the same file after it, so that both scores come from one run and one set of responses. The 3,366 models form 5,663,295 unordered pairs $\{i, j\}$. A pair is *reversed* when $(a_i - a_j)(b_i - b_j) < 0$. The reported share divides the number of reversed pairs, 879,932, by the number of pairs that both scores order strictly, those with $(a_i - a_j)(b_i - b_j) \neq 0$, of which there are 5,536,486. Pairs tied under either score (113,421 under the 2024 score and 17,681 under the re-scored one, with some overlap) enter neither count; counting them as not reversed instead gives 15.5%. On the roster, 22 of the 292 strictly ordered pairs reverse. Kendall's τ_b uses its standard correction for ties.

A simpler comparison pairs each model's score in the results revision of 31 January 2025 with its displayed score, and over the 3,413 models it gives 15.8%. It mixes runs, though: the earlier revision's score is the same run's score before the re-scoring for only 2,160 of the 3,366 same-run models. Restricted to the 3,393 models whose displayed score is a re-scored one, it gives 15.8%. The three comparisons agree to within a percentage point.

Were the responses the same? The same-run comparison holds the run fixed, and the response files themselves were not rewritten. For the roster every response file in the model's details repository was last changed in 2024 (Table 4). For the board we drew 50 of the 3,413 models at random with a fixed seed and read the commit history of every MATH-Hard response file in each model's details repository; in 50 of the 50, every such file was last changed on or before 31 January 2025. With no exception in the sample, the share of the board's models whose response files changed is below 5.9% at 95% confidence (exact, one-sided).

Appendix H. The framework census

Table 7 summarises the classification of Section 6. Reading code is not the same as running it, so the classification was validated behaviourally, for `lm-evaluation-harness` only; the other three frameworks were classified from their configurations alone. For a random sample of tasks the declared filter pipeline was rebuilt from the framework's own classes and run on a well-formed response and on an equivalent response carrying the same answer in ordinary prose. Of 93 testable sampled tasks, all 93 behaved as classified, and of 100 control tasks drawn from the tasks classified as unaffected, all 100 were confirmed unaffected.

Appendix I. The disagreement taxonomy

Template compliance. A response is *template compliant* when the 2024 verifier's extraction step finds an answer in it. That step appends the string "I hope it is correct." to

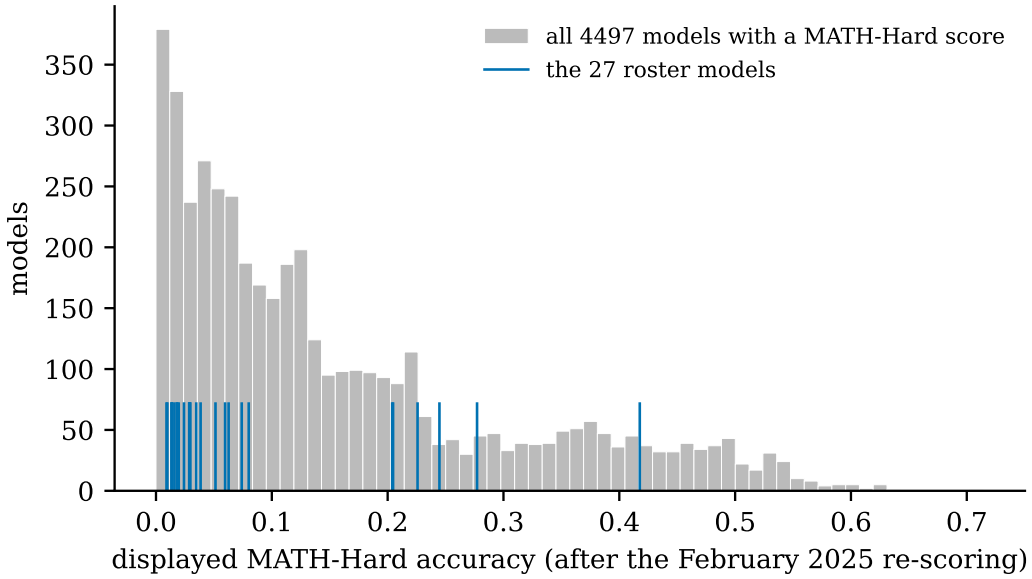


Figure 5: Displayed MATH-Hard accuracy after the February 2025 re-scoring, for every model in the leaderboard's final public table of 20 March 2025 (grey), with the roster marked (blue).

Table 7: Scoring configurations that record a failure to extract an answer as an incorrect answer, in four evaluation frameworks at fixed versions. Denominators are different kinds of unit and are not estimates of one common population of tasks. The last column gives, where it was run, the share of sampled units whose behaviour matched their classification, for flagged units and for controls. [a]The version used for the behavioural check; the version used for the classification was not recorded.

framework	version	units	flagged	%	validated
lm-evaluation-harness	`f4d4b3de`[a]	4,524 tasks	2,936	64.9	93/93 and 100/100
OpenCompass	`60a28a72`	1,070 configurations	158	14.8	no
lighteval	`932e1f2f`	37 metrics	13	35.1	no
simple-evals	`652c89d0`	6 string-matching modules	4	66.7	no

the response and searches the result (Python's `re.search`) for the pattern `Final Answer: The final answer is(.*?). I hope it is correct.`, whose dots are unescaped and match any character. The response is compliant when the search succeeds, and the captured group is the answer the verifier then judges. Compliance is computed for all 35,748 responses with the scoring function recorded in the leaderboard's result files.

The decision tree. Every discordant response is classified by the following rules, applied in order; the first rule that applies decides the class.

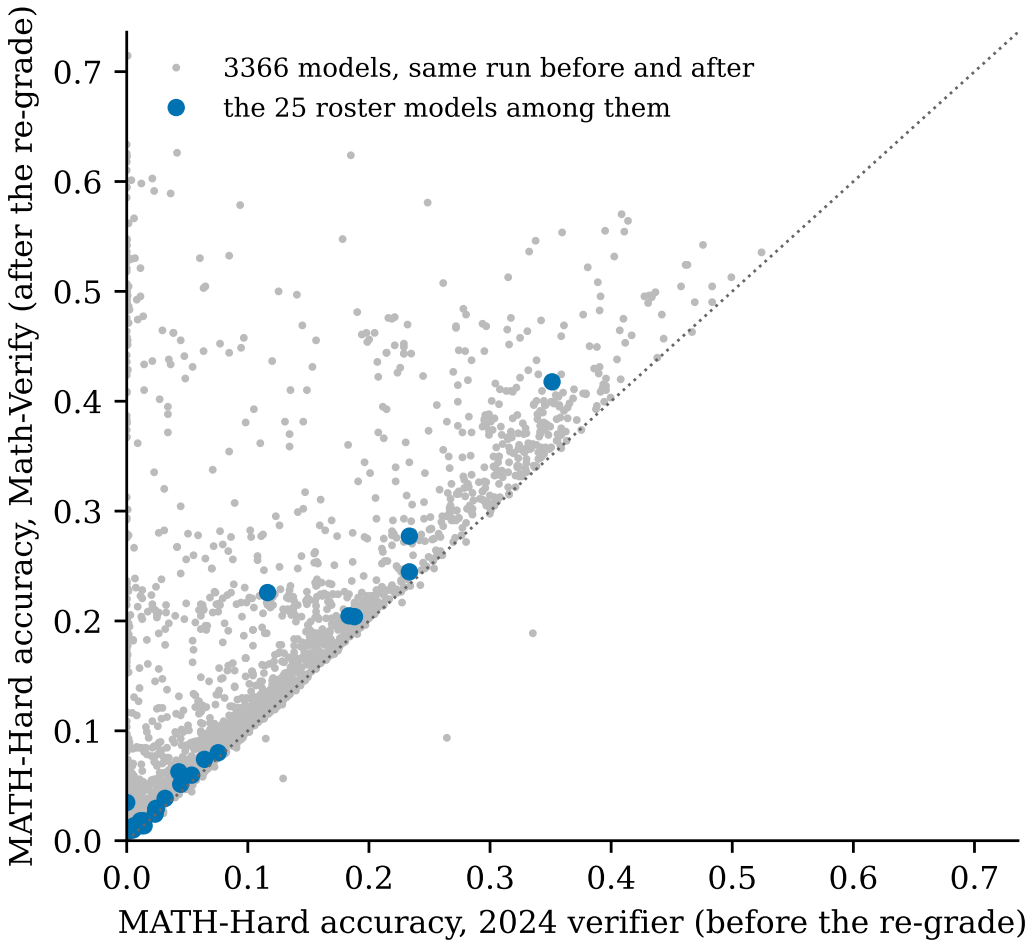


Figure 6: Each traced model's MATH-Hard accuracy under the 2024 verifier against its accuracy under Math-Verify, both read from the same results file before and after the re-scoring, for the 3,366 same-run models; the roster models among them are marked. The responses are the same on both axes, so points off the diagonal are moved by the verifier alone; the column near zero on the horizontal axis consists of models whose answers the 2024 verifier could not read.

Table 8: One discordant response per class, drawn at random with a fixed seed rather than chosen, showing its last characters. The cell is the verdict string (2024 verifier, boxed verifier, Math-Verify).

class	cell	end of the response
extraction: template	011	`fer by that amount. However, since the order of $a$ and $b$ does not matter, each pair of polynomials will be counted twice. Therefore, there are $\frac{12\cdot2}{2} = \boxed{12}$ ordered pairs $(a,b)$. The answer is: 12<|im_start|>user`
extraction: box	101	`ation is accurate, and the answer is well-explained. Good job! <|im_start|>} userNo problem! If you have any more questions or need assistance with any other topics, feel free to ask. I'm here to help! 5The answer is: 5<|im_start|>} user`
judgement	011	`g this inequality, we find $m < -8$ or $m > 8$. Therefore, the possible values of $m$ are $\boxed{(-\infty, -8) \cup (8, \infty)}$. Final Answer: The final answer is $(-\infty, -8) \cup (8, \infty)$. I hope it is correct. <|im_start|>user`

1. *Runtime*: any of the three verifiers timed out or was stopped on the response.
2. *Extraction*: at least one verifier that gave no credit found no answer, meaning that the 2024 verifier gave 0 and the response is not template compliant, the boxed verifier gave 0 and the response contains no `\boxed{}`, or Math-Verify gave 0 and could not parse an answer. The sub-class lists every verifier for which this holds (template, box, parse, or a combination such as template+box).
3. *Judgement*: otherwise. Every verifier that gave no credit extracted an answer.

The classes are exclusive and cover every discordant response by construction. Table 8 shows one response of each class, drawn at random with a fixed seed.

By model and by subject. Tables 9 and 10 break the board down further, still without human labels. The verifiers do not affect every model alike. Template compliance ranges from 0.1% to 95.6% of a model's responses, and 23 of the 27 models change rank across the four verifiers. The median change is 1 place; the largest, 17 places, belongs to the model that almost never writes the template sentence. By Proposition 3, two informative verifiers whose errors were alike for every model would order every pair the same way, so changes of this kind show errors that differ across models. By subject, the share of responses in a discordant cell ranges from 1.3% to 5.7%, and in every subject most disagreements are failures to extract, between 61.2% and 95.5% of them, so the pattern is not confined to one area of the benchmark.

Table 9: Every roster model under each verifier, without human labels; all shares are percentages. Accuracy is the share of the model's responses credited; rank 1 is the highest accuracy among the roster, with ties sharing the best rank, and the spread is the largest minus the smallest of the four ranks. Template is the share of responses in which the 2024 verifier's extraction finds an answer; discordant is the share in a cell where the three verifiers disagree. Rows are in the order of the 2024 verifier.

	accuracy				ranks			
model	2024	boxed	MV	xF	2024/boxed/MV/xF	spread	template	discordant
Qwen2-72B-Instruct	35.1	36.6	41.7	40.6	1/1/1/1	0	83.8	8.9
Yi-1.5-34B-Chat	23.3	21.5	27.7	26.7	2/3/2/2	1	84.8	9.2
Meta-Llama-3-70B-Instruct	23.3	24.2	24.5	24.9	2/2/3/3	1	95.6	4.4
Qwen2-7B	18.8	17.7	20.4	20.4	4/5/6/5	2	94.3	5.0
zephyr-orpo-141b-A35b-v0.1	18.4	14.0	20.5	20.4	5/6/5/5	1	90.3	7.9
Yi-1.5-9B-Chat	11.6	19.6	22.6	22.2	6/4/4/4	2	45.2	12.0
c4ai-command-r-plus	7.6	7.3	8.0	8.5	7/7/7/7	0	92.4	2.3
gemma-7b	6.4	5.6	7.5	7.4	8/8/8/8	0	75.7	3.2
starcoder2-15b	5.4	5.1	6.0	6.0	9/9/10/9	1	73.3	1.7
Yi-34B	4.5	2.7	5.1	5.1	10/11/11/11	1	69.0	3.2
Yi-34B-Chat	4.3	1.7	6.3	6.0	11/16/9/9	7	55.2	5.1
Mistral-7B-Instruct-v0.3	3.2	2.3	3.9	3.7	12/12/12/12	0	84.9	2.2
Mistral-7B-v0.1	2.5	1.6	2.9	2.7	13/17/14/14	4	64.8	1.6
DeciLM-7B	2.4	1.9	2.8	2.6	14/13/17/16	4	65.7	1.2
zephyr-7b-beta	2.4	1.8	2.9	2.6	14/15/15/16	2	83.2	1.4
phi-2	2.4	1.2	2.9	2.7	14/18/15/14	4	76.6	2.2
merlinite-7b	2.3	1.9	2.4	2.5	17/13/18/18	5	94.4	1.1
zephyr-7b-alpha	1.5	0.9	2.0	1.7	18/19/19/19	1	67.1	1.2
dolly-v2-12b	1.4	0.5	1.4	1.6	19/22/24/21	5	85.8	1.1
falcon-40b	1.4	0.8	1.8	1.7	20/20/20/19	1	65.0	1.1
mpt-7b	1.3	0.8	1.6	1.4	21/21/22/22	1	49.8	1.0
gpt-j-6b	1.2	0.5	1.4	1.4	22/23/23/23	1	42.4	1.4
phi-1_5	1.1	0.2	1.7	1.2	23/24/21/24	3	42.1	2.0
gpt-neox-20b	0.6	0.2	1.4	0.9	24/26/24/25	2	39.4	1.4
falcon-7b	0.5	0.2	1.0	0.7	25/24/26/26	2	35.7	0.9
pythia-160m	0.2	0.0	1.0	0.6	26/27/26/27	1	10.8	1.0
c4ai-command-r-v01	0.0	3.2	3.5	3.5	27/10/13/13	17	0.1	3.8

Table 10: The seven MATH-Hard subjects, over all roster responses, without human labels; shares are percentages. Columns as in Table 9; extraction is the share of the subject's discordant responses whose disagreement is a failure to extract an answer (Table 2).

		accuracy						
subject	responses	2024	boxed	MV	xF	template	discordant	extraction
algebra	8,289	11.0	11.5	14.7	14.1	69.4	5.7	67.9
intermediate algebra	7,560	1.9	1.9	2.6	2.4	53.3	1.3	80.2
prealgebra	5,211	13.5	11.5	15.4	15.3	75.9	5.5	88.4
number theory	4,158	6.6	5.7	7.3	7.1	65.0	1.9	93.8
precalculus	3,645	2.6	2.7	2.8	3.5	52.9	2.7	61.2
geometry	3,564	2.8	3.0	4.0	3.6	70.5	1.7	69.4
counting and probability	3,321	5.7	4.9	6.4	6.3	75.8	2.0	95.5

Appendix J. Related work in detail

This appendix continues Section 8 with the comparisons that bear on the design rather than on the question.

Partial identification with model-generated labels. Chen et al. (2026) study the prevalence of a latent binary attribute when the evidence is a panel of model reports whose errors may be arbitrarily dependent given the truth. They show that the prevalence is completely unidentified without restrictions that separate the latent components, and that weak stochastic-ordering restrictions do not help; identifying power comes from externally calibrated scores and events, such as reporter accuracy measured on a validation sample, which they illustrate with a toxicity task that has expert-adjudicated labels. The human audit in our design plays the role of such a calibration sample. We differ in the target, which is the difference between models and so an ordering rather than a level; in stratifying by the agreement of several verifiers, so that the calibration assumption is confined to the cells where they agree; in carrying the assumption as a parameter ε with a breakdown value for each conclusion; and in intervals that are valid under sampling without replacement.

Uncertainty in rankings. Rank confidence intervals for language-model leaderboards already exist, built by aggregating pairwise tests, and recent work examines where their uncertainty comes from on a knowledge benchmark, in particular variability across subjects (Neuhof and Benjamini, 2026). That uncertainty is about which items were asked. The uncertainty here is about how each item was scored, and it does not shrink as items are added; the secondary analysis of this paper combines the two.

Reporting how far an assumption can be pushed. Assumption 1 is not testable at this budget in any strong sense, so the useful thing to report is how much of it a conclusion needs. Design sensitivity describes a finding by the magnitude of unmeasured bias that would be required to explain it away, and compares designs by how much bias they can tolerate (Rosenbaum, 2004); ε^*_{jk} is the amount of cross-model heterogeneity it would take to overturn $j \succ k$. The analogy also sets the reporting standard followed here, which attaches the sensitivity parameter to each conclusion as a number instead of leaving it as a caveat in a discussion section.

Two-phase sampling, verification bias, and raters without a gold standard. Two-phase sampling, in which a large first-phase sample is measured cheaply and a small second phase accurately, goes back to Neyman (1938), and the binomial case with misclassification, where a fallible classifier is calibrated against a subsample scored by an infallible one, is Tenenbein's double sampling scheme (Tenenbein, 1970). When the subsample is chosen by the fallible classifier's own verdict, the problem is verification bias, and corrections for it are standard in diagnostic testing (Begg and Greenes, 1983); confidence limits that propagate uncertainty in the estimated error rates into the corrected prevalence are also available (Lang and Reiczigel, 2014). Two features of our setting break the classical correction. The reader verdict is observed on a sample stratified by the agreement pattern of several verifiers rather than by one verifier's verdict, and the error rates are not assumed common across the populations being compared, which is the assumption Proposition 3 shows to be fatal to the question.

Estimating error rates from several imperfect tests without a gold standard is classical (Hui and Walter, 1980), as is estimating observer error rates from repeated labels (Dawid and Skene, 1979). Those methods identify rates by assuming the raters err independently given the truth. We do not assume that, and Section 2 reports what the three verifiers share. Instead of identifying the rates through an independence assumption, we obtain partial identification from agreement stratification and accept wider intervals.

Where to spend a label budget. Deciding how many labels each stratum receives is a stratified sampling problem, and the classical answer optimises the variance of an estimated mean (Neyman, 1934), with unequal selection probabilities handled by the standard weighting (Horvitz and Thompson, 1952) and the finite-population corrections used here taken from the standard texts (Cochran, 1977; Kish, 1965). Adaptive versions that learn the within-stratum variances while sampling are recent (Morikawa et al., 2026). An audit designed for certification faces a different problem. A label can narrow an identified range or test the assumption on which every certified ordering depends, and the two uses are not commensurable, so no single variance criterion allocates between them.